**An Internship Report**

**on**

# *eLoran* Digital Communication Protocol: Deciphering and Implementation

**by**

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## ABSTRACT

Our world and everyday activities depend on the Global Navigation Satellite Systems (GNSS) for Positioning, Navigation, and Timing (PNT). We know that these satellite systems are insecure and become weak because their signals travel ~20000km. Regarding security vulnerabilities, the ongoing war in Europe has further exposed GNSS's susceptibility to jamming, spoofing, and radio-frequency (RF) signal injection, rendering GNSS unreliable and unsafe. There is an international growing concern regarding GNSS's natural interference and its vulnerabilities, which have led to questions and demands about a resilient PNT provision (Curry, 2014).

**What is a solution?**

Although there exist several national, continental, and worldwide systems, which their signals are quite easy to receive, including DCF77 in Germany, ALS162 in France, MSF in the UK, WWV in the US, JJY in Japan, these systems are dedicated only to timing, having a time accuracy in the milliseconds to microseconds range with a bit rate of 1bit/sec. In 2019, Czaplewski and Weintrit reported that *eLoran*, the enhanced version of Loran-C (Long Range Navigation), is currently the best technical and scientific solution providing effective protection for PNT (Czaplewski & Weintrit, 2019). The Resilient Navigation and Timing Foundation (RNTF) reported that China has developed 294 fibre-based PNT stations with 20-nanosecond accuracy using differential *eLoran*, which is highly reliable and precise (RNTF, 2024). *eLoran* is accurate, available (a terrestrial signal), and it meets the integrity and continuity performance requirements for different PNT applications. It is also a source of time and frequency for applications such as telecommunications (Curry, 2014). This is an independent, dissimilar, and complementary signal to GNSS. People using GNSS can retain security, safety, and communication even when satellite services are disrupted or unavailable (UrsaNav, *ELoran* System Definition and Signal Specification Tutorial, Nov 2011).

As of the time of this writing, *eLoran* can convey several types of messages, including GPS correction information, time information, broadcasting station identifier, short message service, and some other tentative message types (Differential ELoran Reference Station Reelektronika Manual Version 1.0, Release Date, 2007). However, beyond the published standard and online documentation, no open-source implementation of the *eLoran* protocol is available to date. This work aims to bridge the gap by identifying some of the shortcomings in the documentation.

This report discusses the findings about this signal, which focuses on and includes phase extraction and demodulation, soft to hard bit conversion (phase to bit conversion), bits concatenation, cyclic redundancy check (CRC) validation, forward error correction (FEC), and interpretation of bits to human-readable sentences following the standard's message types.

# Chapter 1 – Introduction

## 1.1 *eLoran* vs. GNSS

Enhanced Loran (*eLoran*) is an internationally-standardised PNT service for use in many applications. It is the latest in the longstanding, proven series of low-frequency, **LO**ng-**R**ange **N**avigation (LORAN) systems.

During early World War II, the US developed the Loran-A hyperbolic navigation system operating on a 2 MHz carrier for use at sea, but its application range and accuracy were limited. By late World War II, they had to develop the other Loran systems (B, C, D, F). Loran B was a phase comparison variation of Loran A, including an attempt to redesign the system properly and add new features, since Loran A was rushed and was never very accurate. Phase comparison increased accuracy to about half a cycle (about 0.25 microsecond), a considerable improvement. Loran B also had its transmitters operating in chains instead of pairs, thus allowing simultaneous measurement of two-time differences. Despite Loran B's very precise timing measurements, there were considerable phase distortions at the land-sea interface, resulting in major differences in the phase-to-envelope timing depending on the azimuth from the transmitter. Loran B never made it as a commercial system of navigation due to these issues. Loran-D was a short-range, high-accuracy, low-power tactical system used by the United States Air Force for bombing. It used portable transmitters with a sixteen-pulse transmission instead of eight. Loran-D modified the Loran-C pulse pattern by interleaving an additional 8 pulses after each pulse, leaving a 500 us spacing between pulses as opposed to 1000 uS in Loran C. The system was used for several war exercises in Europe in the 1960's and later. While the U.S. Air Force was developing Loran D, Motorola experimented with the Multi-User Tactical Navigation Systems (MUTNS), a continuously pulsed pseudorandom coded low-frequency navigation system used for drone control. It was sometimes referred to as Loran-F in proposals, but this was not an official designator. Loran-D came out ahead during system evaluations; therefore, no further work was done on MUTNS (Jerry Proc - VE3FAB, 2022).

Loran-C, the successor to Loran-A, came about due to the need for a Loran-like system that could be operated over much greater distances in daylight than Loran-A could provide. The only potential solution was to try Loran techniques at low frequencies, which resulted in experiments that could be received by Loran A equipment. After the Pacific War, there were lots of system decommissioning as the experiments were not followed up on. However, in later years, Loran C became alive again due to some work on phase comparison by a company named Sperry. The US NAVY took interest and recommissioned transmitters that could broadcast a phase coherent 100kHz signal at about 60kW with a wave range of ~3000 miles. Following this, there has been plenty of adoption and development to this day, with power as high as 1.2MW, frequency at 100kHz, and a travelling distance in kilometres. The Loran-C system had a positioning accuracy of 460 meters (US *Federal Radionavigation Systems*, 2005); however, when GPS became available in the 1980s with an accuracy of tens of meters, people began switching from Loran-C to

GPS. The removal of the GPS selective availability (SA) feature by President Bill Clinton in 2000 further improved global coverage. These led to Loran becoming irrelevant. Others kept it as a backup if the GPS failed (ILA, 2007).

*eLoran* evolved from Loran-C due to the vulnerabilities discovered in GPS, as published in the US Volpe report (Volpe, 2001), almost in time with the "9/11" tragedy. The report emphasised the disruption of GPS signals by intentional or unintentional interference, and it further identified Loran as a potential solution to this crucial issue, which further led to an increased interest in modernising the US's Loran systems. In addition to the US Volpe's report, there have been several headlines about possible threats to GNSS's network; these threats remain a reality for the land, marine, and aviation industries. GNSS signals received on Earth are weaker than thermal noise and can therefore be disrupted deliberately or by accident. Space debris collisions with satellites, solar activities, hostile military actions, resulting in blocking GNSS signals, jamming, and spoofing for terrorist and criminal activities, all these and more are threats associated with GNSS.

The US Loran evaluation and modernisation program (Lombardi et al., 2008) resulted in this new enhanced version of Loran with greatly improved performance. *eLoran* meets a set of worldwide standards and operates wholly independently of GPS, GLONASS, Galileo, BeiDou, or any future GNSS (ILA, 2007). It provides the accuracy, availability, integrity, stability and continuity performance requirements for aviation non-precision instrument approaches, maritime port and harbor entrance, approach maneuvers, land-mobile vehicle navigation, location-based services, and is a precise source of time and frequency for applications such as telecommunications, while serving as a reliable backup to GNSS, taking over seamlessly if the satellite signals are lost, thus allowing users to continue to operate and keep systems running. As of today, different countries in the world are now developing their own *eLoran* system. Table 1 below shows a comparison between GNSS and *eLoran* (Yang et al., 2023).

**Table 1.** Comparison of characteristics between GNSS and *eLoran* (Yang et al., 2023).

| **System** | **GNSS** | ***eLoran*** |
|---|---|---|
| Frequency | High (1575.42 MHz) | Low (0.1 MHz) |
| Power | Very Low | Very High |
| Transmission | Space | Terrestrial |
| Jamming | Easy | Very Hard |
| Spoofing | Easy | Very Hard |
| Reach | Global | Continental |
| Accuracy | Best | Good |
| Positioning | 3D | 2D |
| Propagation | Atmosphere | Ground |

## 1.2 *eLoran* evolving from Loran-C

*eLoran* is a modern, digital signal that manifested from the legacy Loran-C system. The primary difference between these two is the addition of a data channel on the already existing Loran-C signal. This data channel can convey different types of messages, including correction information, time information, station ID & health information, short message service, third party information, see Eurofix Message Description and Status in Appendix. This data channel enables *eLoran* to meet the very challenging requirements of highly accurate and signal integrity applications like landing aircraft and bringing ships safely to harbour, even in low visibility conditions (ILA, 2007).

Since the 1970s, with the application of semiconductor batch processing, large-scale electronic manufacturing, microcomputers, and powerful solid-state transmitter technology, the Loran-C system and its broadcasting technology have been and are being improved upon. In 1989, Delft University of Technology in the Netherlands proposed the concept of "Eurofix", which enables encoding data into Loran-C pulses using the Pulse Phase Modulation (PPM) method. Eurofix was adopted by Europe. The US, on the other hand, implemented something called "9th Pulse Modulation" by introducing an extra pulse into Loran-C pulses; this extra pulse carries the data. These two independent forms of data embedding form the Loran Data Channel (LDC) (Lombardi et al., 2008; Yang et al., 2023). The aim was to use Loran-C to broadcast GPS differential correction data and integrity information, forming the prototype for the *eLoran* system. In 2007, the International Loran Association (ILA) released the "Enhanced (*eLoran*) Definition File", which pointed out that the *eLoran* system is a standardised PNT service system (Yang et al., 2023).

Loran-C (effectively *eLoran*) transmissions occur within a chain of transmitters, a master, and some secondaries. All *eLoran* transmitters use modern solid-state transmitters (SSX) and control technology. The time and frequency control systems of the transmitters are designed for *eLoran* operation, and they apply phase corrections in a continuous manner. The timing of each transmitter station is generated from a local unit of three Caesium standard clocks that are themselves synchronised periodically to UTC, by comparison with available master standards (*International Atomic Time (TAI), a time scale calculated at the Bureau International des Poids et Mesures (BIPM) in France, using data from some 400 atomic clocks in over 70 national laboratories*). By doing this, the transmissions are locked to UTC and so provide a source of UTC-trackable timing, totally independent of GNSS, so-called "*sky free UTC*" (Curry, 2014). When a station is noticed operating outside the standard, in case of serious and harmful loss of synchronisation, it is taken off the air right away to ensure that receivers immediately cease to utilise its signals. *eLoran* has stopped using blinking to indicate stations transmitting test signals; integrity messages are now conveyed through the LDC (UrsaNav, 2011).

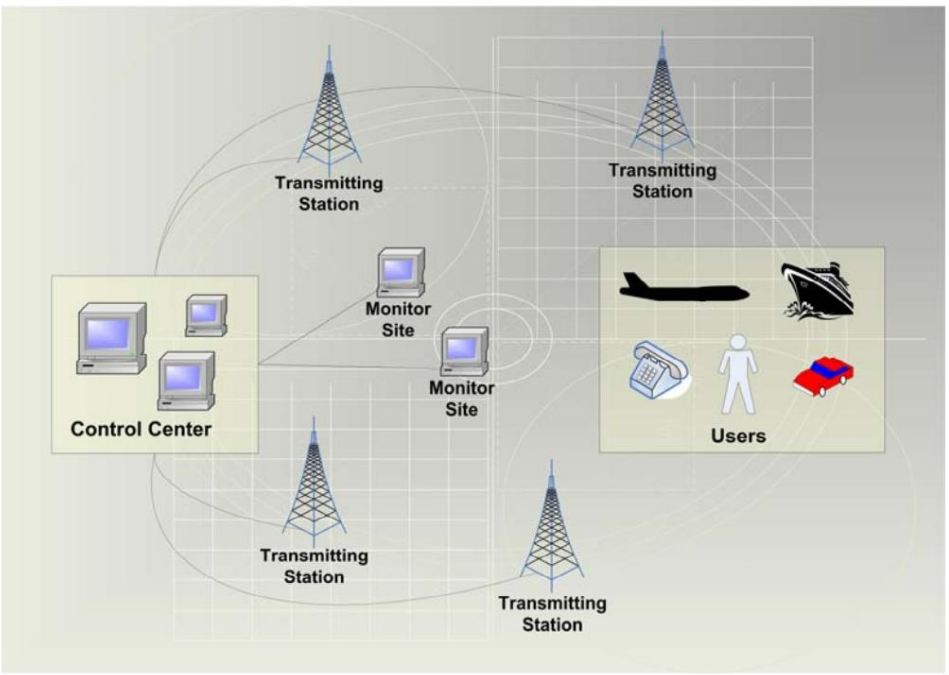
In addition to the transmission stations running unattended, sufficient personnel are present at control centres to respond rapidly to issues while maintaining the published very high levels of availability and continuity (ILA, 2007).

Along with transmission stations and control centres, monitor sites, located in the *eLoran* coverage area, are used to provide integrity for the user community. The receivers used at monitor sites constantly keep track of the *eLoran* signals, providing real-time information to control centres concerning signals in space. Some of the monitor sites are often used as reference stations to generate the data channel messages. Selected sites also have a minimum of one highly accurate clock for synchronisation to UTC to provide time and frequency corrections for timing users. A monitoring network could also be present to provide real-time maritime differential corrections and provide warnings for aviation users (ILA, 2007). See Figure 1 below for the *eLoran* system concept.

*eLoran* continues using the existing Loran-C transmitters to deliver PNT information. These transmitters operate at a frequency of 100 kHz, radiating precisely timed pulses in hundreds of kilowatts of power. To deliver highly precise navigation, the pulses must be timed with an accuracy of nanoseconds. Because of this, they can fulfil the additional function of distributing precise time over long distances (Curry, 2014).

These low-frequency transmissions can penetrate buildings, which makes it possible to receive the signal indoors with a magnetic-field antenna, "*H field antenna.*" (Czaplewski & Weintrit, 2019). This potential has been thoroughly assessed in two of Chronos's UK research projects: GAARDIAN and SENTINEL (Sentinel, 2014; Chronos, 2017).

**Figure 1.** The *eLoran* system concept (UrsaNav, 2011).



Recent studies and tests indicate that *eLoran* meets or exceeds the accuracy, availability, integrity, continuity, and coverage requirements vital to achieve 8- to 20-m maritime Harbour Entrance Approach (HEA) and aviation Required Navigation Performance (RNP) for non-precision approaches (Lombardi et al., 2008). It also allows absolute UTC to be recovered with an accuracy of 50 nanoseconds, as well as meeting the stratum 1 frequency standard needed by telecommunication users (ILA, 2007).

*eLoran* has proven to be reliable; it retains the powerful long-range (not microwatts and microwaves like GNSS) and unjammable low-frequency signals (90-110 kHz) of its predecessor. This means that small, low-cost, *eLoran* receivers, even built into GNSS units, can mitigate the impact of disruptions to GNSS. *eLoran* is the culmination of technical developments undertaken worldwide over the past two decades, which now meet the very demanding requirements of 21st century users (ILA, 2007). France, the United Kingdom, and South Korea are working together on strategic plans to standardise *eLoran* and to establish an international consultative body for *eLoran* cooperation (RNTF, 2025). The New York Stock Exchange in the United States has tested the use of the *eLoran* system as a backup for the GPS for time synchronisation. South Korea is testing the use of a differential *eLoran* system as a backup for GPS for precision navigation in ports. China has also added *eLoran* timing systems in similar financial institutions and communications fields to improve the security of critical infrastructure (Offermans et al. 2017).

Despite the strengths of the *eLoran* signal, there are quite a few factors that could limit its accuracy. These are due to its propagation path, space weather, and local electrical interference. The signal propagates as ground-wave (land and sea) and sky-wave (sky, ionosphere). The electrical conductivity of seawater is quite high, greater than that of land or mountains. This affects the speed of propagation, resulting in signals over sea arriving earlier at a receiver than those over land. If a signal travels wholly or partly over land, it will experience an additional delay with respect to its propagation over seawater; this delay is called the Additional Secondary Factor (ASF), directly affecting the time of arrival of signals conveying UTC service. Since *eLoran* now conveys data, there are provisions to provide corrections to this difference as a message type (Curry, 2014).

In addition to the groundwave signals, a sky-wave component is also radiated, reflected by the ionosphere, and returned to the earth at night. This unwanted component could interfere with the useful groundwave signal, as it would be, sky-wave signals arrive later at the receiver when compared to the groundwaves. *eLoran* receivers are designed to make measurements on groundwave pulses before any skywave interference arrives, rendering them insensitive to the variations of the ionosphere and the influence of space weather events (Curry, 2014).

Electrical interference from frequencies around 100kHz can be generated by consumer devices, which can reduce the signal-to-noise ratio of the received *eLoran* signals, especially indoors. Since these interferences

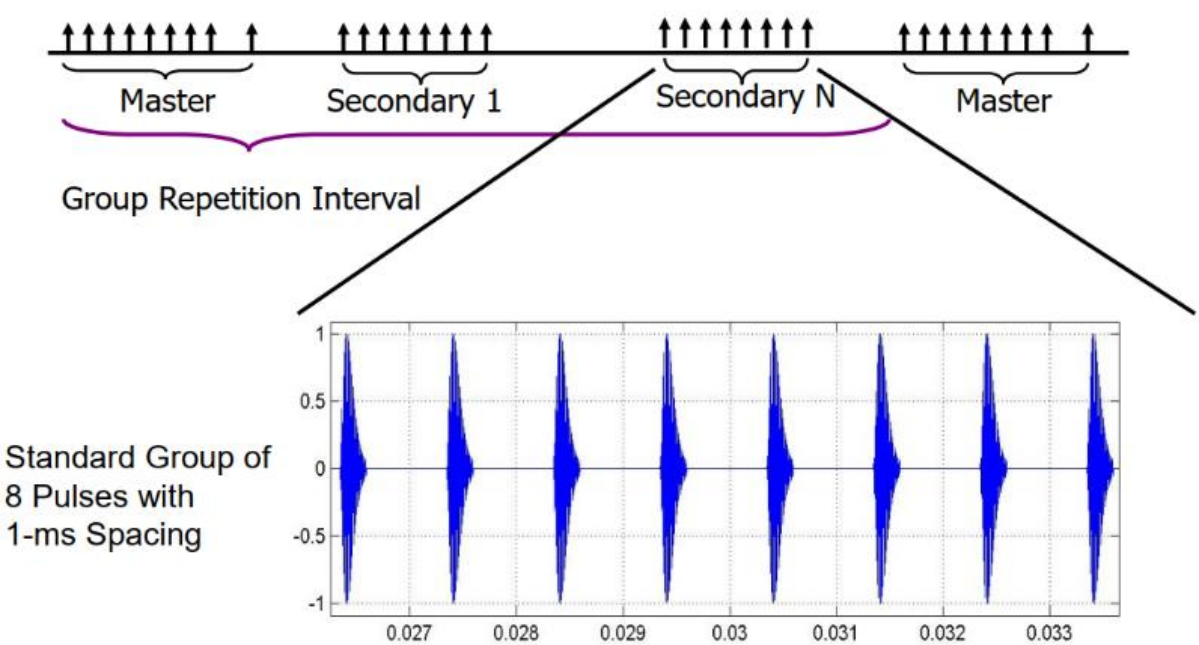
are localised, the effect could be intense. A combination of filtering, choice of antenna position, and orientation are ways to mitigate these interferences (Curry, 2014).

### 1.3 *eLoran* signal structure

The *eLoran*, a high-precision groundwave-based pulsed signal, adopts the Loran-C system, with a carrier centre frequency of 100 kHz in the 90-110 kHz band, having power in hundreds and thousands of watts. Loran-C, which was primarily for PNT purposes, has 8 pulses separated by 1 millisecond, with an additional pulse separated by 2 milliseconds to identify the master station, so the master station comprises 9 pulses with pulses 1 to 8 each spaced by 1milliseconds, pulses 8 and 9 are spaced by 2 milliseconds, and the secondary stations comprise 8 pulses, each separated by 1 millisecond. These pulses, either master or secondary forms a group. Pulses 1 and 2 are used for PNT since Loran-C. *eLoran* became a thing by introducing a data channel on other pulses, preserving the first two pulses for the existing PNT purpose as specified by the International Telecommunication Union (ITU) Recommendation M.589-3 (ITU, 2001).

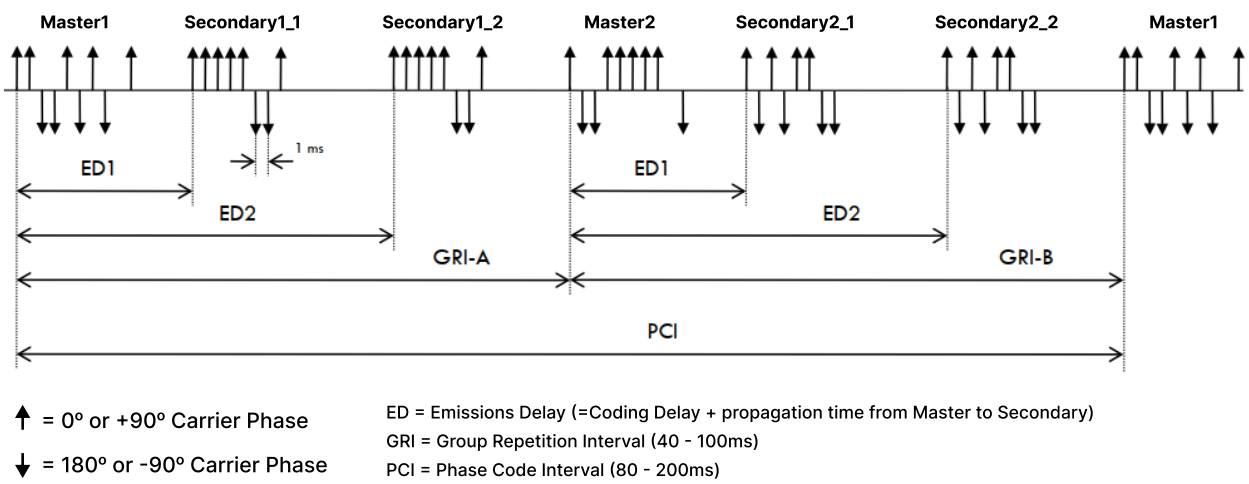
For *eLoran* transmissions, the transmitters that belong to the same *eLoran* system form a chain. An *eLoran* chain is identified by its group repetition interval (GRI). To explain what GRI is, we start with what we already know: a group is made up of 8/9 pulses, which could be from a master or secondary station. In a chain, there is one master and up to 5 secondary stations, with each station in a respective chain transmitting a group of pulses (8/9). This translates to 5 or 6 groups, depending on the number of stations in that chain; however, each pulse groups are transmitted one after the other, first with the master and then secondary(ies). After all secondary pulse groups are transmitted, a GRI is formed. A GRI is the time between the first pulse of a master and the first pulse of the next master. See Figure 2 below for clarity.

**Figure 2.** *eLoran* pulses and group repetition interval (UrsaNav, 2011).



In addition to GRI, there are other terms like emission delay (ED) and phase code interval (PCI), which will be explained shortly. The *eLoran* carrier signal at 100kHz for a GRI is phase coded either 0° or 180°, which could also be ±90°; this enables master/secondary identification by comparing the received phases with the standard. Emission delay is the time difference between the first pulse of a master transmission and the first pulse of a secondary transmission, or the interval of time in microseconds (µs) between the start of transmitter n's time slot and the start of the first transmitter's time slot (SAE 9990, 2018). A phase-coded interval is essentially two GRIs; that's a simple way to define it. Technically, it is the time difference between the first pulse of a master transmission and the first pulse of another master transmission, with the same carrier phase code.

**Figure 3.** *eLoran's* EDs, GRI, PCI, and Phase Codes (UrsaNav, 2011).




More phase codes are provided by the Systems Management Standard - SAE International (SAE 9990, 2018). Several countries are broadcasting *eLoran* signals, each chain having its own GRI. This list is not exhaustive, as we could only provide this information based on available resources; some are listed below.

- The United Kingdom, GRI = 6731.
- China South Chain, GRI = 6780.
- The Kingdom of Saudi Arabia, GRI= 8830.
- The Republic of South Korea, GRI = 9930.

### 1.4 The Loran Data Channel (LDC)

The most important difference between legacy Loran-C and *eLoran* is the addition of a Loran Data Channel (LDC) to the transmissions. The LDC brings a highly robust, long-range channel that carries digital data messages. Although the LDC has a low bit-rate, it has enough capacity to deliver these digital and high-priority messages to users. The initial purpose of these messages was tied to PNT, including:

- to convey differential GPS corrections, similar to the ones in other DGPS systems.

- to ascertain the correct and safe operation of the transmission to users, thereby ensuring high navigation integrity.
- to convey corrections for the small temporal changes of signals' timing received in some harbours, where the very highest location accuracy is required (Curry, 2014).

Over the years, there have been several improvements in *eLoran*, and the message types were not left out. As of 2001, according to the ITU in the M.589-3 document, up to 16 messages were planned, with three message types active, and the rest reserved (ITU, 2001). Two implementations of LDC exist, namely;

- Eurofix standardised by ITU.
- 9th Pulse Modulation.

### 1.5 Eurofix

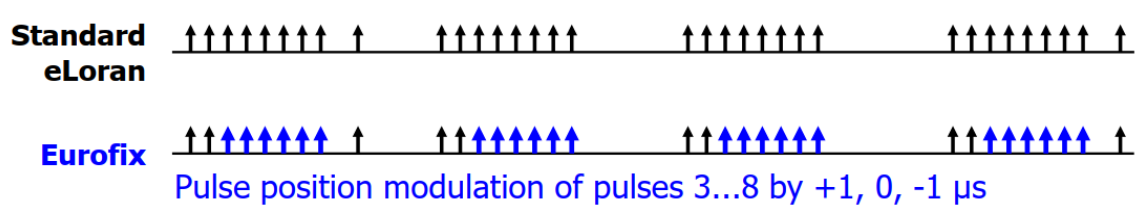
Eurofix was proposed as a concept in 1989 at Delft University of Technology in the Netherlands, which implements LDC utilising the Pulse Position Modulation (PPM) method. Ever since then, Eurofix has evolved and become an LDC standard internationally, recognised by the Radio Technical Committee for Maritime Services (RTCM), documented by the International Telecommunication Union (ITU, 2001), with its messages defined in a document entitled "*Eurofix Message Format*"; the current version of which, ver.2.15, is dated March 2014 (Offermans, 2014) (Czaplewski & Weintrit, 2019).

Europe adopted Eurofix as its LDC implementation of *eLoran*. In the UK, the General Lighthouse Authority (GLA) considered *eLoran* to provide an alternate/backup source of UTC to the telecommunications industry. It concluded that *eLoran* can cushion the impacts of GNSS failures, and by using both systems, it can have a robust network infrastructure (Lombardi et al., 2008). The UK sees *eLoran* as an official time source, similar to the MSF time station that broadcasts on 60 kHz. Countries, including the UK, China, the Kingdom of Saudi Arabia, and others not mentioned but presumed, upgraded their legacy Loran-C systems to support *eLoran*, implementing Eurofix as the LDC.

Eurofix has the following properties;

- Its physical layer is Loran-C with a carrier frequency of 100kHz.
- Tri-state Pulse Position Modulation (3S-PPM).
- Pulses 3-8 are 3S-PPM modulated in each pulse group. See Figure 4 below for illustration.

**Figure 4.** Pulses 3 to 8 modulated for Eurofix.

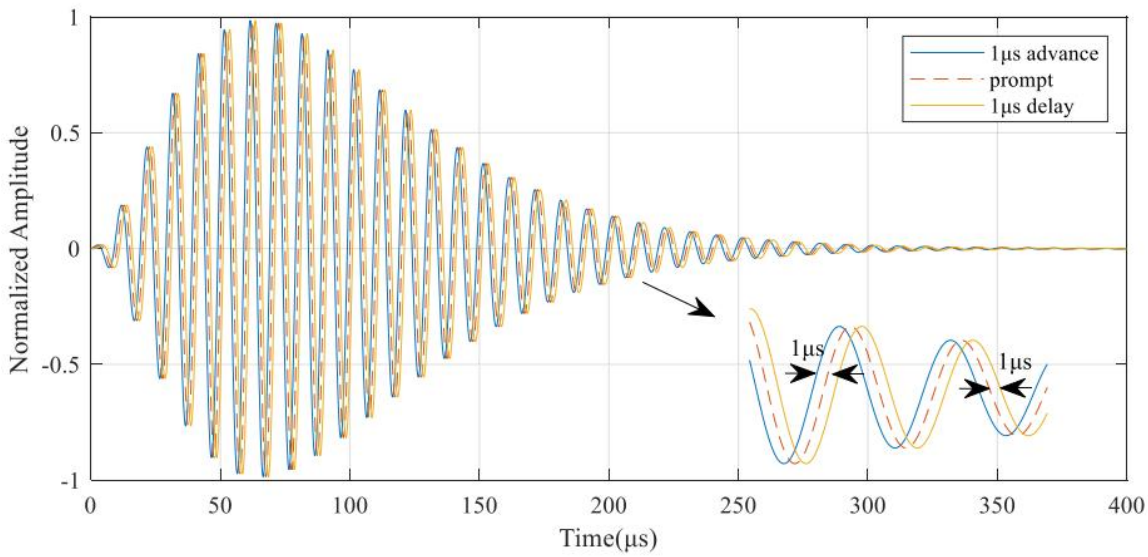


- The modulated signal has a time shift of 1 microsecond relative to the unmodulated pulse.
- The three possible modulation states (Advanced, Prompt, and Delayed pulses) are shown below.

**Table 2.** States of modulation (ITU, 2001).

| Pulse State | Transmission time minus time of reference pulse (µs) | Indication |
|---|---|---|
| Advanced Pulse | -1 | - |
| Prompt Pulse | 0 | 0 |
| Delayed Pulse | +1 | + |

**Figure 5.** 3S-PPM in a plot (Zhang et al., 2024).




- The number of advanced and delayed pulses in a pulse group should be equal and must cancel out.
- The modulation of six pulses in a pulse group yields 141 possible balanced patterns (+, 0, -).
- Of these 141 balanced patterns, 128 patterns represent data, one pattern indicates no data transmission, and 12 patterns should not be used; see Appendix for patterns.
- The pattern "0 0 0 0 0 0" is used to indicate that data is not transmitted.
- Each pattern is of six characters derived from the 3S-PPM of 6 pulses.
- Each pattern represents a 7-bit binary block of data; see Appendix for the pattern/data translation.
- 30 GRIs (30 consecutive pulse groups) form one message.
- 30 GRIs of 7 bits per message = 210 bits = one message.
- Depending on the GRI, the bandwidth could range from 70-175bps = 1.2 – 3 seconds per message.
- The frame consists of a message type (4 bits), a message body (56 bits), a cyclic redundancy check (CRC) (14 bits for data integrity), and Reed-Solomon parity (140 bits for forward error correction (FEC)). See Appendix for more information.

- There are 16 possible message types defined, with 8 having been assigned to existing/new services (Offermans, 2014), Eurofix Message Description and Status in Appendix.
- The CRC is generated by using the following polynomial;

$$G(x) = x^{14} + x^{13} + x^7 + x^5 + x^4 + 1$$

- As regards FEC, a systematic Reed-Solomon (RS) (30,10) $2^7$-ary code is applied to all messages.
- All messages consist of 30 symbols, each symbol = 7-bit element = 1 pattern = 6 PPM pulses.

  Note: do not confuse patterns with symbols, patterns are generated from the 6 PPM pulses (indicated with +, 0, -), each pattern then translates into a symbol (decimal or hexadecimal), see Appendix – pattern/data translation for clarity.
- The symbols are elements of the Galois Field GF(128) constructed using the primitive polynomial.

$$p(x) = x^7 + x^3 + 1$$

- The symbol "0" corresponds to a 7-bit value of 127.
- The FEC parity is defined with the generator polynomial below;

$$g(x) = \prod_{i=1}^{20} (x - \alpha^i)$$

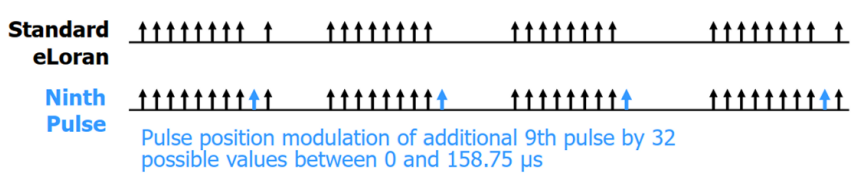
- The first transmitted pattern of the FEC-encoded message is the least significant symbol of that message.
- Eurofix messages are transmitted consecutively without interleaving.
- After the last pattern of a message is transmitted, the next transmitted pattern is the first pattern of the next message; this means that there is no start or stop indication of a message in the transmissions (ITU, 2001).

During the course of this work, although some of the properties listed above were quite ambiguous (due to inadequate documentation and examples, as opposed to other existing and known protocols), you will read in later chapters all the different methods applied to break and decipher this protocol and achieve success.

### 1.6 9th pulse Pulse Position Modulation (PPM)

This type of modulation was designed to have little to no impact on the operational signal to avoid violating the existing signal specification or interfering with the legacy Loran-C. In this modulation scheme, an additional pulse is inserted in time (1000 µs) following the eighth pulse in a pulse group, which effectively makes master transmission 10 pulses and secondary transmissions 9 pulses (SAE 9990-2, 2018).

**Figure 6.** Ninth pulse PPM in a GRI (UrsaNav, 2011).



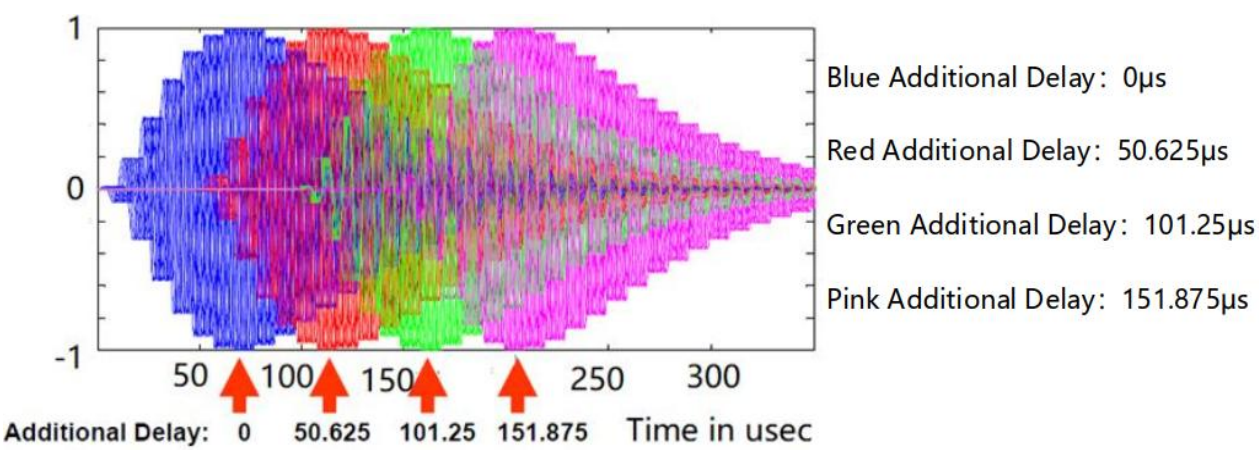
9th pulse PPM has the following properties (Lombardi et al., 2008; SAE 9990-2, 2018; Yang et al., 2023);

- Its physical layer is Loran-C with a carrier frequency of 100kHz.
- It uses a 32-state PPM to encode data.
- Each signal position is defined by a 5-bit symbol; the first two signals are roughly delayed, and the last three digits indicate a small delay. This way, the data transmission rate is 5 bits/GRI.
- The phase coding of the 9th pulse is the same as the previous navigation pulse.
- The zero-symbol offset is 1000 µs after the eighth navigation pulse.
- The delay time of the remaining 31 symbols relative to the zero symbol is shown in microseconds.
- The formula below gives the ideal delays. The actual delays are the ideal values shifted to coincide with the ticks of a 5 MHz clock, see Appendix for symbol delays from zero-symbol offset (µs).

$$d_i = 1.25 mod\,(i, 8) + 50.625\, floor(i/8)$$

- A message is 120 bits in length. Since a GRI can only convey 5 bits, a message = 24 GRIs with a maximum time length of ~2.4 seconds.
- A 120-bit message comprises three components: a 4-bit message type, a 41-bit message payload, and a 75-bit parity component for correction using RS encoding, see Appendix for components.
- There are 16 possible message types, and only 4 have been identified:
    - Reference Site Phase Corrections
    - Almanac Messages
    - Government Use Only
    - Station ID and Time of Day. All shown in message types under Appendix.

**Figure 7.** The ninth pulse modulation given in (Yang et al., 2023).




As far as this study is concerned, it is not beneficial to describe the 9th pulse modulation further at this point, because the work was centred on Eurofix. This modulation type is implemented in North America, in the Canadian East Chain with (GRI - 59300 µs) and the North Pacific Chain (GRI - 99900 µs) (Lombardi et al., 2008).

In this report, Chapter 2 describes acquisition (online software-defined radio), phase correction, phase extraction, phase to pattern matching, bit concatenation, CRC calculation approach to find a match & start of a message, and forward error correction. Chapter 3 discusses the results, similarities/discrepancies/under documentation of the standard, with respect to our findings. We conclude this report with key conclusions based on this work and publicly available information. See you in the next chapters.

# Chapter 2 – Methods

Open-source, widely available, and free utilities were used to carry out this work, including Linux, GitHub, Octave, KiwiSDR, Brute Force CRC, and Libfec. This choice was based on already existing work (kiwiclient, Libfec, bruteforce-crc) and on challenges with acquiring the signals. We will now explore a step-by-step process of how the *eLoran* signal was deciphered.

## 2.1 Acquisition

This work took place in 2025/2026 in Besancon, France. As of the said date, there is no *eLoran* transmission station in France; the closest is in Anthorn, Cumbria, in the United Kingdom. Aside from this setback, we do not possess the *eLoran* receivers or other acquisition setup that meets the *eLoran* standard. The solution used was an online software-defined radio known as KiwiSDR.

### 2.1.1 KiwiSDR on the Web

This is a software-defined radio that runs on the SeeedStudio BeagleBone Green single-board computer. It is a device that can be purchased to serve as a radio or even run an operating system. This hardware will not be discussed further because we did not use it. The interesting feature of Kiwi is its over 900 publicly available SDR receivers online; the physical hardware is not needed in this case. A map pulled from the Kiwi's website showing all SDR receivers in the world is shown below.

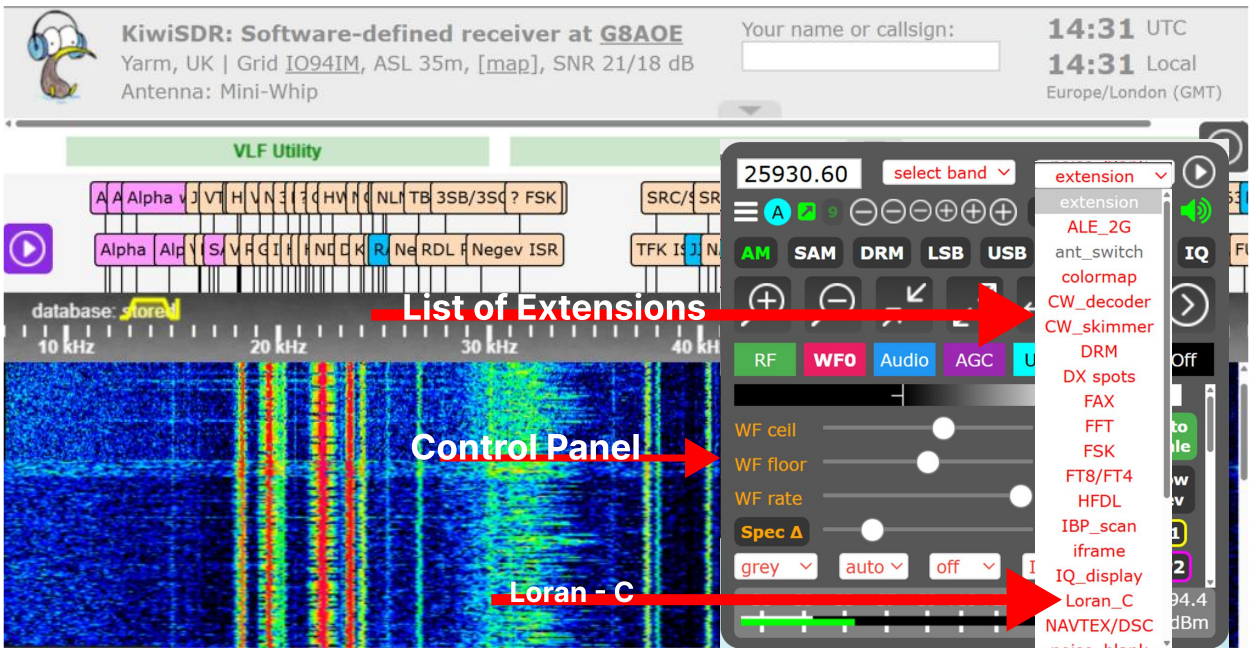
**Figure 8.** Kiwi SDR receiver locations @ http://rx.linkfanel.net/ or http://map.kiwisdr.com/

As regards *eLoran*, we know where *eLoran* transmission stations are, in the UK, Saudi Arabia, China, and Korea. With this information and this map, we basically select the closest KiwiSDR receiver to an *eLoran*

station. By hovering on a preferred receiver on the map, a small window appears showing the band, location, antenna, number of connected users, and signal-to-noise ratio (SNR) of the receiver.

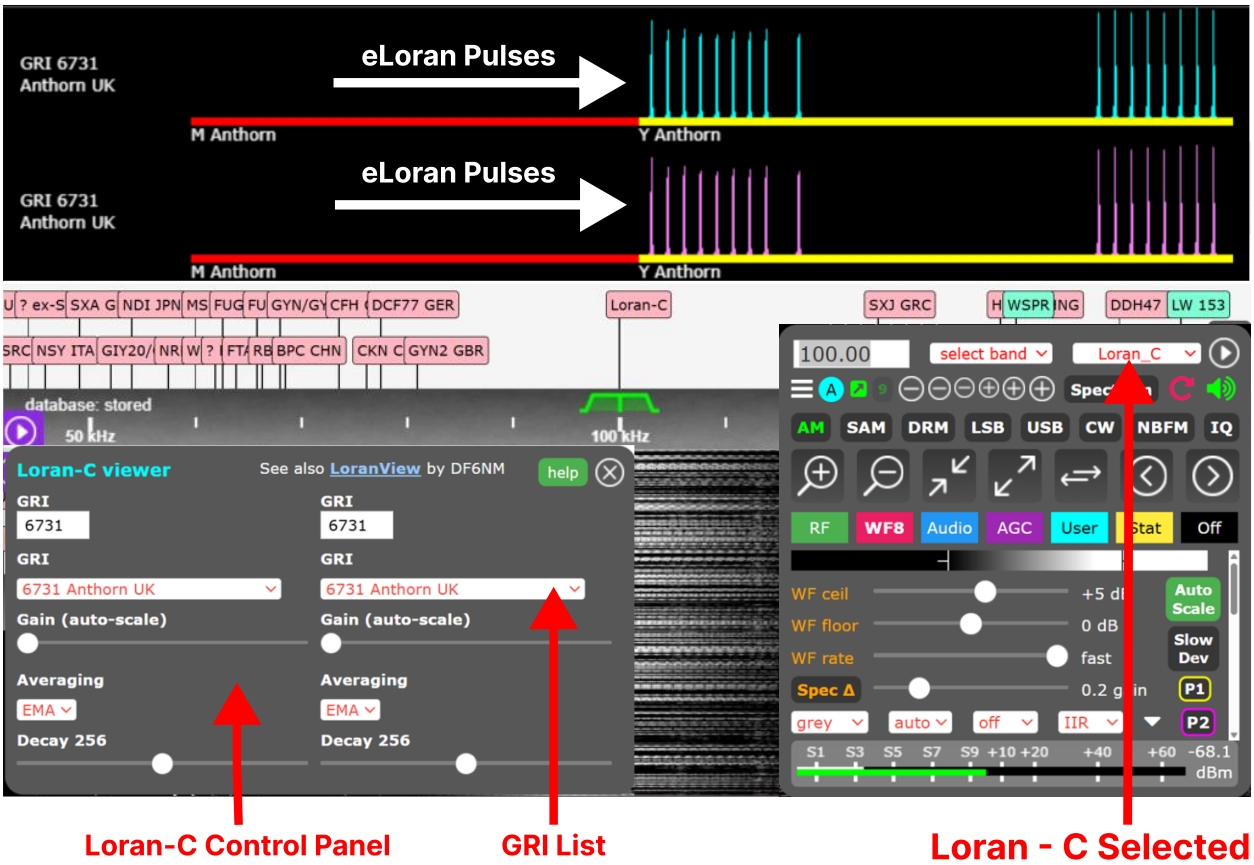
Clicking on a receiver opens a webpage (similar to all other receivers), as shown below in the figure. On this webpage, Kiwi integrates extensions, which can be selected at the top right of the control panel. These extensions are noise blank, noise filter, waterfall, FFT, IQ display, Loran-C display, WSPR decoder, etc. More are in development, according to Kiwi.

**Figure 9.** KiwiSDR receiver web interface (KiwiSDR).




Clicking on Loran-C on the extension list opens another control panel on the left, specifically for Loran-C, which allows one to select from a list of different stations based on GRI or allows one to input the GRI. The figure below is from a receiver in Penrith, UK, very close to Anthorn. Anthorn is listed in the GRI, and we can see the station's pulses in the top-right corner of the page.

**Figure 10.** *eLoran* pulses from Anthon on the KiwiSDR webpage.




This webpage interface allows us to see the pulses and verify that a particular *eLoran* chain is in operation. We can see master pulses and all secondary pulses in a chain. In addition, we can assess the signal's strength, cleanliness, or noise level as received by the receiver before proceeding with the actual automated acquisition.

### 2.1.2 Kiwiclient - https://github.com/jks-prv/kiwiclient

This is the Python client with several utilities for KiwiSDR, which allows one to issue commands to the KiwiSDR, receive data streams with audio samples, IQ samples, S-meter, and waterfall data. After viewing the signal on the webpage, with one of the kiwiclient utilities (kiwirecorder.py), we automated the reception of the signal's IQ samples as a WAV file (we started with Anthorn). It is worth mentioning that the receiver, on the fly, performs frequency demodulation, stripping off the 100kHz carrier signal, and then attaches GNSS timestamps to the IQ samples.

**Using Kiwiclient.**

- **Dependencies**: Linux or macOS, make, Python3, NumPy, Octave.
- **Cloning**: git clone the repository; https://github.com/jks-prv/kiwiclient. In the *Makefile*, there are examples about how to use the utilities, which can be checked using the command "*make help*".

- **Recording**: we recorded samples using the following command;

  ***./kiwirecorder.py -d . --station=G0GHK --tlimit=900 -s g0ghk.uk -p 8050 -f 100 -w -m iq***

  ***./kiwirecorder.py*** - this is the recorder located in the kiwiclient directory.

  ***-d .*** - This is the output directory where the recorder saves the wav file. We saved in the current directory using dot (.)

  ***--station=G0GHK*** - this is the station name; the argument could be anything of choice. For proper identification, we used the station names as written on the kiwi map.

  ***--tlimit=900*** – this option (*--tlimit*) specifies the record time in seconds. 900 seconds = 15 minutes.

  ***-s g0ghk.uk*** – this option (*-s*) specifies the source, which is the domain name or IP address of the receiver we previously viewed on the web page.

  ***-p 8050*** - this option (*-p*) specifies the port number of the service. This port number is usually part of the address on the web page of a receiver.

  ***-f 100*** – this option (*-f*) represents the frequency of the signal to be acquired in kHz. We specify 100 (in kHz) as the argument because Loran-C has a 100kHz carrier signal.

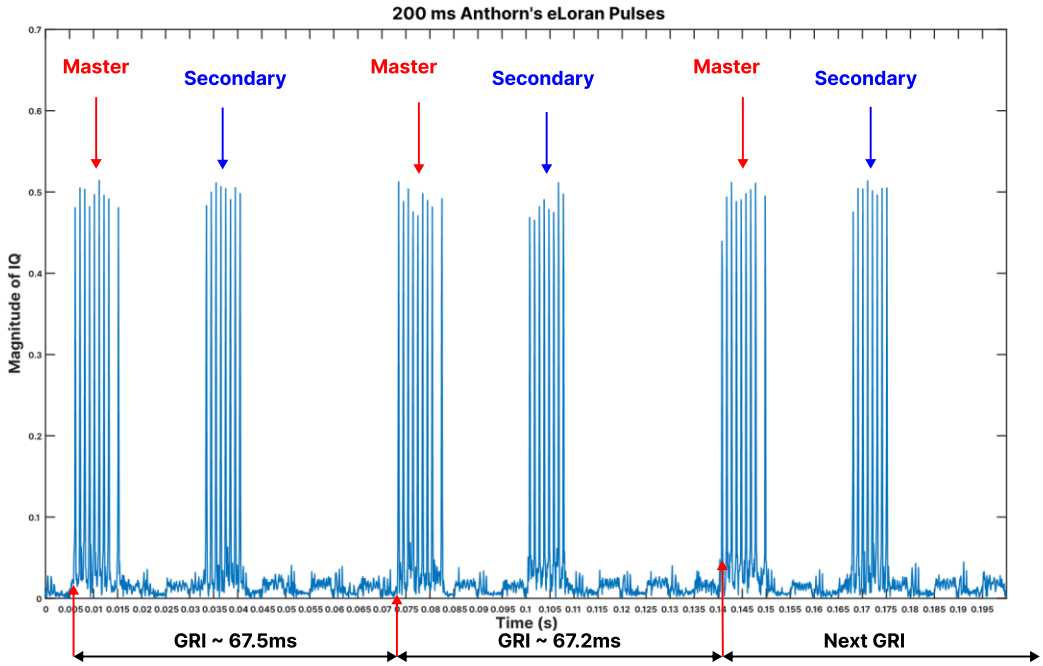
  ***-w*** – this option (*-w*) tells the receiver to write a wav file in the output directory stated.

  ***-m iq*** – this option (*-m*) specifies the mode of record; we passed *iq* because we wanted the signal's IQ samples.

- **Output:** *kiwirecorder.py* produces a WAV file and a text file containing the receiver's web address, port number, longitude, and latitude.
- **Working with WAV files:** Kiwi also provided some Octave files to read the WAV files recorded by the KiwiSDR receivers.
  - An Octave extension, *read_kiwi_iq_wav.cc* compiled with "*make install*" produces an Octave (*.oct*) library.
  - An Octave function that calls the compiled Octave library: *proc_kiwi_iq_wav.m*. The wav file is passed as input to the function. The function extracts IQ samples and GNSS time stamps for each sample (*x* and *xx*), receiver's sampling rate (*fs*), and the oldest GNSS position solution (*last_gpsfix*).
- **Implementation:** Below are extracts from the script where *proc_kiwi_iq_wav.m* is called, and its outputs are assigned to variables z and t. A 200ms portion of the IQ sample's magnitude is also extracted (z50ms) and plotted.

```
[x,xx,fs,last_gpsfix]=proc_kiwi_iq_wav(dlist(l).name);
z=cat(1,xx.z);z=z(floor(fs/5):end);z50ms=z(1:floor(fs/5));
t=cat(1,xx.t);t=t(floor(fs/5):end);t=t-t(1);t50ms=t(1:floor(fs/5));
plot(t50ms,abs(z50ms))
```

**Figure 11.** 200ms plot of Anthorn's *eLoran* signal (magnitude (IQ) vs time).




From the plot above, in the Anthorn chain, there is only one secondary pulse group per GRI. We see that the pulses do match the standard with 1ms spacing, master pulses being 9, and secondary pulses being 8. We can also confirm that the GRI (~ms) on the plot matches closely with Anthorn's GRI (67.31 ms).

We briefly observe the phases to check whether they align with the standard's phase codes, as shown in Figure 3 above. The phase codes can be denoted with + for +90° and - for -90°, just as in the figure below. This would be applied when comparing the received IQ samples phases with the standard.

**Figure 12.** Phase code representation.

| Master1 | Secondary1_1 | Secondary1_2 | Master2 | Secondary2_1 | Secondary2_2 | Master1 |
|---|---|---|---|---|---|---|
| ++--+-+- + | +++++--+ | +++++--+ | +--+++++ - | +-+-++-- | +-+-++-- | ++--+-+- + |

In addition to this reminder, we also know that Eurofix uses 3 pulse position phase modulation for data that corresponds to ±1us (±36°) for advance/delayed or 0° for prompt. This translates to six pulse positions relative to the carrier phase (+90°,+90°±36°) and (-90°,-90°±36°). A portion of the script to extract and plot the phase, in addition to figures showing the individual pulse phase, and an exploded view are below.

```
kmag=find(abs(z50ms)<(max(abs(z50ms))/th));
z50ms(kmag)=NaN;
plot(t50ms,180/pi*angle(z50ms),'Linewdith', 1.5);hold on
k=find(angle(z50ms)>=0);
```

```
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi)
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi+36)
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi-36)

k=find(angle(z50ms)<0);
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi)
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi+36)
line([min(t50ms)max(t50ms)],[mean(angle(z50ms(k))) mean(angle(z50ms(k)))]*180/pi-36)
```

**Figure 13.** 200ms Anthorn *eLoran* pulses and phases.

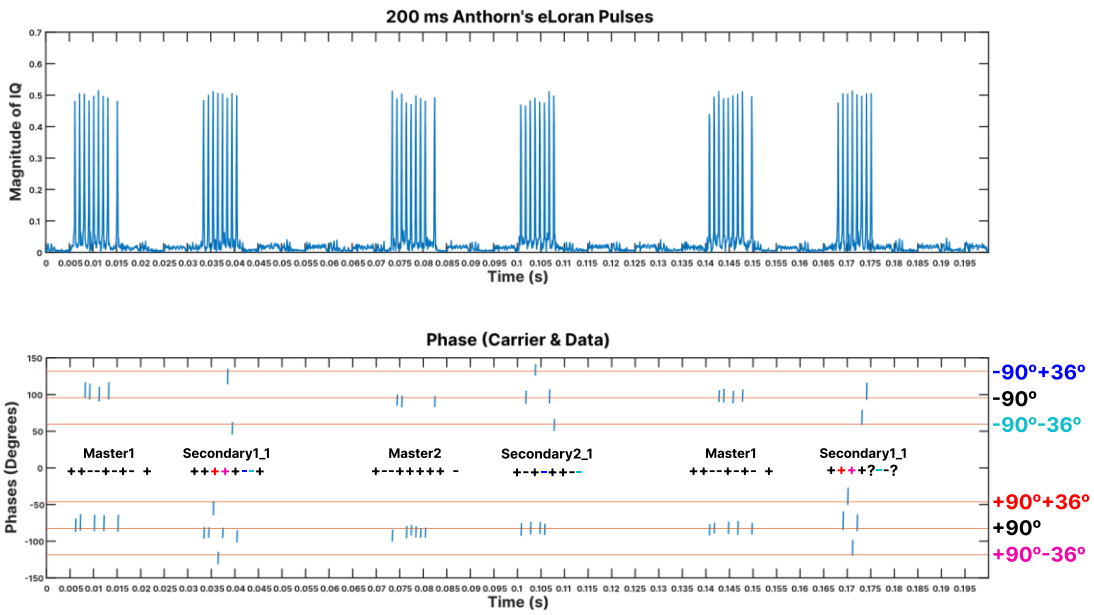
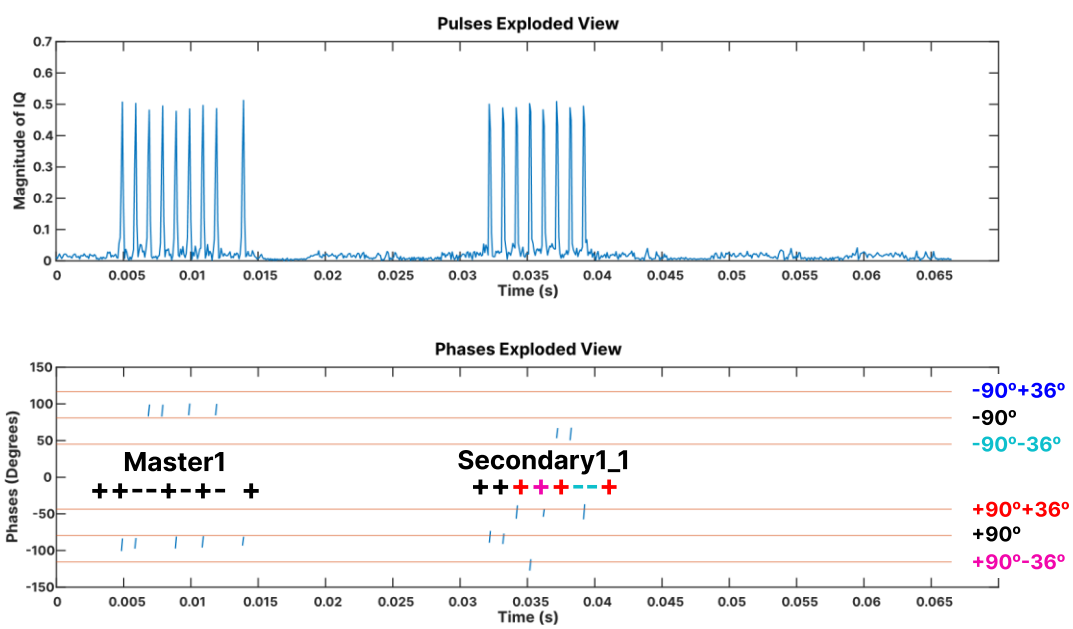



**Figure 14.** 200ms Anthorn *eLoran* pulses and phases – Exploded view.



As we hoped, the phase codes matched the standard. The master pulses do not carry data because their phases were all either ±90 degrees. The secondary pulses conveyed data, due to the ±1 µs PPM, visible as ±36 degrees, as seen in the plots.

## 2.2 The phases

Having established the above, since the signal is a baseband signal, down-converted and filtered by the kiwiSDR, the next step was to extract the phases throughout the WAV file, though this was done in chunks (per pulse group).

### 2.2.1 Temporal segmenting and pulse detection

We start by taking some GRIs to identify the start of the next strong pulse group, which was done by comparing the signal envelope (the 5 GRIs) against a fraction of its maximum value divided by a user-defined threshold, acting as an energy detector without considering the phase. Once the start of the next strong pulse group is identified, it is adjusted backwards by half a pulse width to align with the burst's leading edge. The algorithm avoids false detections by ensuring that the new pulse group occurs at an expected interval, which is GRI-dependent. If the condition fails, the algorithm raises a burst position error. See script lines below for achieving this.

```
kinittmp=find(abs(z(kinit:kinit+5*GRI*fs-1))>max(abs(z(kinit:5*GRI*fs+kinit-1)))/th); %identify start of strong pulse
kinit=kinit-1+kinittmp(1)-floor(dk/2); % start of strong pulse adjusted
tinitold=tinit; tinit=t(kinit);
if ((abs(tinit-tinitold-0.04)>0.005) && (abs(tinit-tinitold-(GRI-0.04))>0.005) && (tinit>.1)) % false detection condition
printf("%f burst position error\n",tinit);end % burst error for false detections
```

*eLoran* pulses are spaced 1ms from each other, with 9 master pulses and 8 secondary pulses. Once the start is identified, an 11ms window, enough to capture a pulse group (master or secondary), is specified. The complex data of this window is then extracted.

```
zuseful=z(kinit:kstop); %complex data extraction for 11ms window
```

Inside this window, the peak magnitude of each pulse is checked by iterating from pulses 1 to 9. The script ensures that pulse group overlap is avoided by restricting the magnitude search to a time frame. The results, which are sample indices of the pulse amplitude peaks, are stored for further processing.

```
[~,pos(m)]=max(abs(zuseful(k+(m-1)*dk:k+m*dk-dk/2))); % sample indices of the pulse amplitude peaks
```

To ensure 1ms spacing between pulses and 2ms spacing between pulses 8 and 9 for the master, differences in pulse position are checked to validate the pulse group structure. Whenever an interval greater than ±1ms (except for 2ms in master) is detected, the script prints a "*pulse position error*".

```
if (isempty(find(abs(posdiff-1)>(2/fs*1000)))==0) % check for 1 ms or 2ms pulse spacing
printf("%f: pulse position error: ",tinit);…… % error is raised
```

### 2.2.2 Phases extraction

We proceed by extracting the phases, which comprise the carrier phase, phase offsets (propagation offsets and receiver local oscillator offsets), coarse phase, and fine phase.

After validating a 1ms/2ms time gap between pulse positions, the carrier phase of each pulse is extracted directly as the argument (angle) of the complex sample at the amplitude peak with the formula/command below;

*ph=arg(zuseful(pos));*

$$\mathbf{ph = arg[zuseful(pos)] = arctan\left(\frac{Q[zuseful(pos)]}{I[zuseful(pos)]}\right)}$$

This extracted carrier phase is further processed by subtracting the mean phase across the 8/9 pulses to remove phase offsets induced by propagation phase shifts and the receiver oscillator's offset, thus leaving the coded and data-modulated phases. This step is shown in the lines below.

```
phmean=mean(ph); % taking mean of phase
ph0sbefore=[ph0sbefore ; phmean];
t0before=[t0before ; tinit];
ph-=phmean; % subtracting the mean phase across the 8/9 pulses
```

*eLoran* signals are phase-coded with 0° or 180° (also ±90°) for master/secondary identification and for skywave rejection. After mean subtraction, the phases fall within two groups and are classified as positive and negative: phase > 0 as positive and phase <= 0 as negative. Since Eurofix modulates data by ±36°/0°, a phase shift is carried out for easier subsequent processing, discarding the coarse phase.

```
kphasepos=find(ph>0); kphaseneg=find(ph<=0); % first find all the positive and negative
ph(kphaseneg)+=pi/2; ph(kphasepos)-=pi/2; % THEN phase shift
```

### 2.2.3 Station identification

With the positive and negative phases, station identification is possible. Using the standard phase codes, we classify pulse groups as either master/secondary, into a GRI, and further into a PCI. The algorithm performs this identification by checking whether a particular pulse (based on the pulse index) belongs to the positive or negative group, labelling it as master/secondary, either "+" or "-", and categorising them into GRIs (1 or 2)

### 2.2.4 Fine phase and soft-hard bit conversion.

Data is phase-modulated with ±36°/0° onto pulses 3-8; this is Eurofix. The next step involved extracting the fine phase. From section 2.2.2, the phases have been brought to a common phase range (adding and subtracting *pi/2*), where only a 36° phase change with respect to phase values close to zero remains.

In this script, we find the phases > +18° and represent them as either +1 or -1, and the phases < -18° are represented as either +1 or -1. When the phase is close to/equals 0°, it is represented as 0. Do not be confused, all the possibilities were tested, as there is no information regarding this conversion. Depending on the phase angle, the hard bits are concatenated into two different variables, thereby forming a pattern.

```
% identify fine phase (+ with +/-36 degrees for 0/-1/+1 or - with +/-36 degrees)
khard=find(ph>+36/180/2*pi);bitpos(khard)=+1;bitneg(khard)=-1; % soft bit to hard bit:
khard=find(ph<-36/180/2*pi);bitpos(khard)=-1;bitneg(khard)=+1; % soft bit to hard bit:
```

According to the standard, the hard bits in the pattern must cancel out; this means we must have an equal number of +1 and -1. The script ensures this and raises an error when this condition is not met.

### 2.3 Pattern/data translation and bits concatenation

The standard provides a pattern-symbol translation table, see Appendix. Having extracted the hard bits into patterns above, we look up each pattern in the table to get the corresponding decimal or hex value and assign the result to two variables.

```
[~,respos]=ismember(bitpos(3:8)',pattern,'rows');   % concatenate hard bits into symbol
[~,resneg]=ismember(bitneg(3:8)',pattern,'rows');   % concatenate hard bits into symbol
```

Moving forward, the script converts the decimal/hexadecimal number into binary, stores and concatenates them into four possible arrays.

```
binresposrl=[mod(respos,2) binresposrl]; % least significant bit (newest) to the left
binresnegrl=[mod(resneg,2) binresnegrl]; % least significant bit (newest) to the left
binresposlr=[binresposlr mod(respos,2)]; % least significant bit (newest) to the right
binresneglr=[binresneglr mod(resneg,2)]; % least significant bit (newest) to the right***
```

### 2.4 Finding the start/stop of a message – CRC

According to ITU-R M.589-3, messages should be transmitted consecutively without interleaving. The pattern transmitted in the first pulse group after the last pattern of a message shall be the first pattern of the next message (ITU-R M.589-3, 2001). There is no sync word in the messages. Intuitively, it is possible to use the CRC of a message to find its beginning and its end. In addition to intuition, the radio data system (RDS) specification for VHF/FM highlights that a continuous stream of bits requires synchronisation, and sliding CRC calculation until a match is detected is one way of detecting the beginning of sentences in the absence of a synchronisation word (EN50067_RDS_Standard, 1998). With the above in mind, following the Eurofix message components arrangement and number of bits (type, body, CRC, FEC comprising 210 bits, see Appendix), we find the start and stop of a message by running a sliding CRC search on the bit stream from section 2.3. Eurofix has a 14-bit CRC in bits 57 to 70; more details in the Appendix.

### 2.4.1 CRC

A cyclic redundancy check is a known data integrity technique to check if a payload has been altered or not during transmission. The CRC of a message is computed using a polynomial division (XOR); the remainder of this division is the checksum, which is attached to the message. In addition to the polynomial and the remainder, there are other terms, including Bit width, Initialisation (Init), Reflected Input (RefIn), Reflected Output (RefOut), and Final XOR Value (XorOut), and an optional check value. Below are the Eurofix CRC properties as stated in the standard.

- Name: CRC-14, 14-bit in width.
- Polynomial: $G(x) = x^{14} + x^{13} + x^{7} + x^{5} + x^{4} + 1 = 1\ 1\ 0\ 0\ 0\ 0\ 0\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1 = 0x60b1$
- Initialisation, Reflected Input, Reflected Output, Final XOR Value: not stated in the standard.

There exist different types of CRC implementations, including normal (discard or keep MSB/LSB), reflected (discard or keep MSB/LSB), Koopman, and reciprocal; all these CRC implementations were tried, but did not yield a positive result.

In addition, we discovered a CRC utility on GitHub (https://github.com/nitram2342/bruteforce-crc) that can slide through a large bit-stream and brute-force the checksum based on the options attributed to its command. This utility produced so many possible checksum values, making it difficult to find the right checksum.

We have to mention that the standard does not provide enough/tangible information to compute the CRC for *eLoran*; however, we had a breakthrough.

We decided to try all possibilities by altering the orientation of the payload, the polynomial, and the remainder. The algorithm loads the bitstream, counts the first 56 bits (payload), computes the CRC for this payload, and then compares the result with the next 14 bits. If the computed checksum matches the 14bit checksum in the bitstream, the script jumps to the next 210th bit; however, if no match is found, it slides with 1 bit and runs the process again, while also changing the orientations. This process acted on all four bitstream files. The last bitstream *binresneglr* yielded the result. It is important to mention that this success allows us to know the right soft-hard bit conversion, bit concatenation, bit orientations, and checksum orientation. The lines below show the working orientation and the CRC computation.

```
data56 = fliplr(window70(1:56)); % the payload
g = [1 1 0 0 0 0 0 1 0 1 1 0 0 0 1]; % 0x60b1 – the polynomial
frame(i:i+14) = bitxor(frame(i:i+14), gen); % CRC XOR computation
rem14 = fliplr(frame(57:70)); % Last 14 bits, the checksum.
```

### 2.5 Forward Error Correction (FEC)

*eLoran* uses Reed-Solomon's forward error correction to correct corrupted symbols. Let's remember that a symbol is a 7bit number generated from 6 patterns generated from 6 pulses. If one of the pulses is interfered with, the whole pulse group gets corrupted, thereby corrupting the symbol. RS does not correct bits, but symbols. Ideally, this RS step should be done before CRC to make corrections where necessary, but because CRC is easier to work with (finding message start/stop and giving us some hope & positivity) and not as computationally intensive as RS, it was carried out first. The two *eLoran* implementations (Eurofix and 9th Pulse Modulation) employ the RS Code, although with different correcting capabilities. Since this work is focused on Eurofix, its RS Code is decoded and reported.

Eurofix uses a systematic Reed-Solomon (RS) (30,10) $2^7$-ary code. This translates into 30 codeword symbols (N) per frame/message, 10 data symbols (K) per frame/message, and 20 parity symbols (N-K). RS (30,10) can correct a maximum of 10 symbols out of the 30 symbols based on the RS standard correcting capability.

$$(N-K)/2$$

Below are some peculiarities of Eurofix RS;

- All symbols belong to a Galois field of $128 = 2^7$
- These symbols are generated with a primitive polynomial; $p(x) = x^7 + x^3 + 1 = 137 = 0x89$
- It is specifically stated that the symbol (dec/hex) "0" must correspond to a 7-bit of 127 (127 in binary).
- As opposed to the widely known relationship between alpha (α) and the binary value in an RS code, Eurofix specifies the opposite, that the binary number (b) be considered the value of the power of alpha (α) as below;

$$\text{Usual} = b_6\alpha^6 + b_5\alpha^5 + \ldots\ldots + b_0\alpha^0$$
$$\text{Eurofix} = \alpha^{b6\ldots\ldots b0}$$

- The FEC parity is defined by the following generator polynomial;

$$g(x) = \prod_{i=1}^{20}(x - \alpha^i)$$

- Lastly and importantly, the first transmitted pattern of an FEC-encoded message should correspond to the least significant symbol of that message. This means the parity symbols are transmitted first, followed by the dataword (payload, CRC) symbols.

We rely on already existing work (*libfec*) @ https://github.com/quiet/libfec to make FEC corrections. Based on the above Eurofix RS specifics, the algorithm extracts the 210 bits for a message from the bitstream (we

know because of the sliding CRC search) and converts them to symbols. These symbols and other parameters, like the primitive polynomial and symbol size in bits, are passed to *libfec* as input. *libfec* outputs the corrected symbols (codeword) if errors were present. Do note that if the number of erroneous symbols is greater than ten, the correction will not be possible. It is also worth mentioning that a sliding FEC algorithm can be used to find the start/stop of a message without using CRC, although it is a computationally intensive workload that takes more processing power and time.

### 2.6 Messages

With the sliding search, we were able to identify blocks of messages, since they are 210 bits long. The script jumps to the next 210th bit when there is a match. In addition to positive results, a concern is the number of false positives that occurred, believed to be due to the cyclic nature of CRC (we are still investigating this). We disregard these false positives by converting a message into a human-readable sentence. If it is a valid message type, we jump to the next message that is 210 bits away. If it fails to be a valid message, we check the next match. Below is a figure showing a portion of matches, including false positives, of an Anthorn recording with file name ***20251014T122009Z_100000_G0GHK_iq***.*wav* in this GitHub directory. The Kiwi recorder names the recorded WAV files automatically.

- **20251014**: Date of recording, 14th of October, 2025
- **T122009Z**: Location of receiver
- **100000**: Recorded signal's frequency
- **G0GHK**: User-defined station name in kiwi recorder's command
- **iq**: Type or mode of recording.

**Figure 15.** CRC sliding matches and false positives.

```
*** VALID CRC MATCH *** at bit 14
Start bit index: 14 (window 14..83)
Data (56 bits)  : 01100101101011011100001110011011010011010101111100101100
Computed CRC(14): 01101110110000
Tx CRC (14)     : 01101110110000
Full 70-bit seg : 0011010011111010101100101101100111000011101101011010011001101110110000
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 15
Start bit index: 15 (window 15..84)
Data (56 bits)  : 00110010110101101110000111001101101001101010111110010110
Computed CRC(14): 11011101100000
Tx CRC (14)     : 11011101100000
Full 70-bit seg : 0110100111110101011001011011001110000111011010110100110011011101100000
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 225
Start bit index: 225 (window 225..294)
Data (56 bits)  : 00000011011000000000000111001110011010111110001000100110
Computed CRC(14): 00010001001111
Tx CRC (14)     : 00010001001111
Full 70-bit seg : 0110010001000111110101100111001110000000000001101100000000010001001111
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 226
Start bit index: 226 (window 226..295)
Data (56 bits)  : 00000001101100000000000011100111001101011111000100010011
Computed CRC(14): 00100010011110
Tx CRC (14)     : 00100010011110
Full 70-bit seg : 1100100010001111101011001110011100000000000001101100000000100010011110
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 435
Start bit index: 435 (window 435..504)
Data (56 bits)  : 00110010110101101110000111001111001100010001010010010110
Computed CRC(14): 00101001001111
Tx CRC (14)     : 00101001001111
Full 70-bit seg : 0110100100101000100011001111001110000111011010110100110000101001001111
-----------------------------------------------------------------------------------------
```

```
*** VALID CRC MATCH *** at bit 642
Start bit index: 642 (window 642..711)
Data (56 bits)  : 00011011000000000000111001111111101100100011100100110010
Computed CRC(14): 00001100000010
Tx CRC (14)     : 00001100000010
Full 70-bit seg : 0100110010011100010011011111111001110000000000001101100000001100000010
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 643
Start bit index: 643 (window 643..712)
Data (56 bits)  : 00001101100000000000011100111111110110010001110010011001
Computed CRC(14): 00011000000100
Tx CRC (14)     : 00011000000100
Full 70-bit seg : 1001100100111000100110111111110011100000000000001101100000011000000100
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 644
Start bit index: 644 (window 644..713)
Data (56 bits)  : 00000110110000000000001110011111111101100100011100100110 0
Computed CRC(14): 00110000001001
Tx CRC (14)     : 00110000001001
Full 70-bit seg : 0011001001110001001101111111100111000000000000110110000000110000001001
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 645
Start bit index: 645 (window 645..714)
Data (56 bits)  : 00000011011000000000000111001111111101100100011100100110
Computed CRC(14): 01100000010010
Tx CRC (14)     : 01100000010010
Full 70-bit seg : 0110010011100010011011111111100111000000000000110110000000110000010010
-----------------------------------------------------------------------------------------
*** VALID CRC MATCH *** at bit 646
Start bit index: 646 (window 646..715)
Data (56 bits)  : 00000001101100000000000011100111111110110010001110010011
Computed CRC(14): 11000000100100
Tx CRC (14)     : 11000000100100
Full 70-bit seg : 1100100111000100110111111110011100000000000001101100000001100000100100
-----------------------------------------------------------------------------------------
```

The matches were further cleaned to keep only those that are 210 bits apart, see crc_eloran.txt. See Figure 16 below showing a portion of these messages.

**Figure 16.** Cleaned messages separated by 210 bits.

```
binresneglr: 0015 data 0110100111110101011001011011001110000111011010110100110 0
binresneglr: 0225 data 0110010001000111110101100111001110000000000011011000000
binresneglr: 0435 data 0110100100101000100011001111001110000111011010110100110 0
binresneglr: 0645 data 0110010011100010011011111111001110000000000011011000000
binresneglr: 0855 data 0110100110011110110111010000101110000111011010110100110 0
binresneglr: 1065 data 0110010000110101000000011000101110000000000011011000000
binresneglr: 1275 data 0110100101111011101000100100101110000111011010110100110 0
binresneglr: 1485 data 0110010010001000110100001100101110000000000011011000000
binresneglr: 1695 data 0110100111000010000010111100101110000111011010110100110 0
binresneglr: 1905 data 0110010001101110101010010010101110000000000011011000000
binresneglr: 2115 data 0110100100010101010110101010101110000111011010110100110 0
```

According to the Eurofix message format V2.14 (Offermans, 2014), see Appendix for Eurofix message description and status, there are sixteen message types, with only 8 allocated. In this work, we focused on message type 4 (station ID/health message) and type 6 (UTC message) because we are able to fact-check them; the remaining six message types are correction messages, short message service, and third-party data. Some of these facts include recording date, year, latitude and longitude, leap seconds, and hour of the year.

### 2.6.1 Message Type 4 – Eurofix Station ID/Health Message

Eurofix message type 4 contains the reference station id and the health of the Eurofix reference station. The message type 4 payload is described below;

- starts with 4 bits to identify the message type
- continues with 10 bits for the reference station id with a range of 0 - 1023
- Next are 3 bits for reference station health, with a range of 8 states
- A system indicator is represented as 2 bits
- Master or secondary identification takes 3 bits with 8 possible stations
- A subtype with 2 bits for either longitude, latitude, or no position information.
- 32 bits represent degrees in units of 1e-7 degrees.

### 2.6.2 Message Type 6 – Loran UTC Message

Message type 6 contains the information on the Loran-C UTC provision. The messages are divided into 4 subtypes, of which 2 are filled in. The message type 6 payload is described below;

- starts with 4 bits to identify the message type
- a message subtype with 2 bits, message subtype 1 is further described;
    - 29 bits for the time at Master/Secondary in hours, with a 10us unit of range 5368 secs (> 1 hour)
    - Hour of the year is represented with 14 bits, unit is 1 hour, range is 16384 hours (> 1 year)
    - 6 bits represent Year, with a range of 64 years (2000 – 2063)
    - A spare 1 bit.
- message subtype 2 (2 bits) is further described;
    - 29 bits for the time at Master/Secondary in hours, with a 10us unit of range 5368 secs (> 1 hour)
    - Precise time takes 10 bits, unit in ns, a range of 10.24 us (> 10 us)
    - Leap seconds (TAI-UTC) take 9 bits, unit is 1 sec, range of -256 to 255 secs
    - 2 bits represent Leap second change, unit is 1 sec, range is -1 to 1 sec.

What we also discovered is that the analysis of these messages is not a direct application of the Eurofix message format. Each component of the 56-bit payload had to be flipped/reflected before decimal conversion. You will see more in the results section.

In conclusion to chapter 2, this work is documented at https://github.com/jmfriedt/kiwisdr_timetransfer, and a Readme file is available that describes the scripts. This *libfec*-based RS decoding has been wrapped as Octave libraries generated with "*make -f Makefile.rs*" to produce *rs30_10_decode.oct* (and *rs30_10_encode.oct* for validation purpose but not used). The main files are *processeloran.m*, *crc_eloran.m* and *rs30_10_decode.oct*.

*processeloran.m*: takes a WAV file as input, extracts and converts phase to bits, and outputs bitstreams.

*crc_eloran.m*: takes output of *processeloran.m* and finds the start/stop of a message by sliding through the bitstream, computes the checksum of 56 bits, compares this calculated checksum with the transmitted checksum (14 bits next to 56 bits), and reports if there is a match.

*rs30_10_decode.oct*: if *crc_eloran.m* does not find a match, it checks if there are 140 bits behind the 56 bits, calls *rs30_10_decode.oct* to make corrections.

# Chapter 3 - Results and Discussions

We are quite happy to show and discuss the results achieved in this work. We were indeed able to read the messages from some stations, while data from others proved difficult.

Several *eLoran* signals were recorded (from Anthorn, Qatar, Korea, and South China) with KiwiSDR. We viewed the signal on the receiver's website before attempting to record, which allowed us to see how strong and clean the signal was. Kiwi does not allow opening a receiver webpage, and at the same time, record from the recorder with Python; only one connection is allowed from an IP address. With the Kiwi GitHub provisions, we could extract the signal's complex data and receiver sampling frequency, which is always 12kHz. The complex data was further processed, phases were extracted and converted to hard bits, bits were concatenated, the CRC sliding method was used to find messages and validate checksums, and *libfec* was used to correct message symbols where errors were not more than ten. The CRC sliding technique did yield interesting results, some false positives, and helped to establish the following properties, which the standard failed to provide;

- Soft to Hard bit conversion:
  - +36° corresponds to the advanced pulse indicated as -1 (-)
  - -36° corresponds to the delayed pulse indicated as +1 (+)
- We concatenated the bits with the newest to the right.
  - The CRC sliding algorithm reflects the input (56-bit payload)
  - The polynomial is used as is.
  - The calculated checksum (reminder) is reflected.

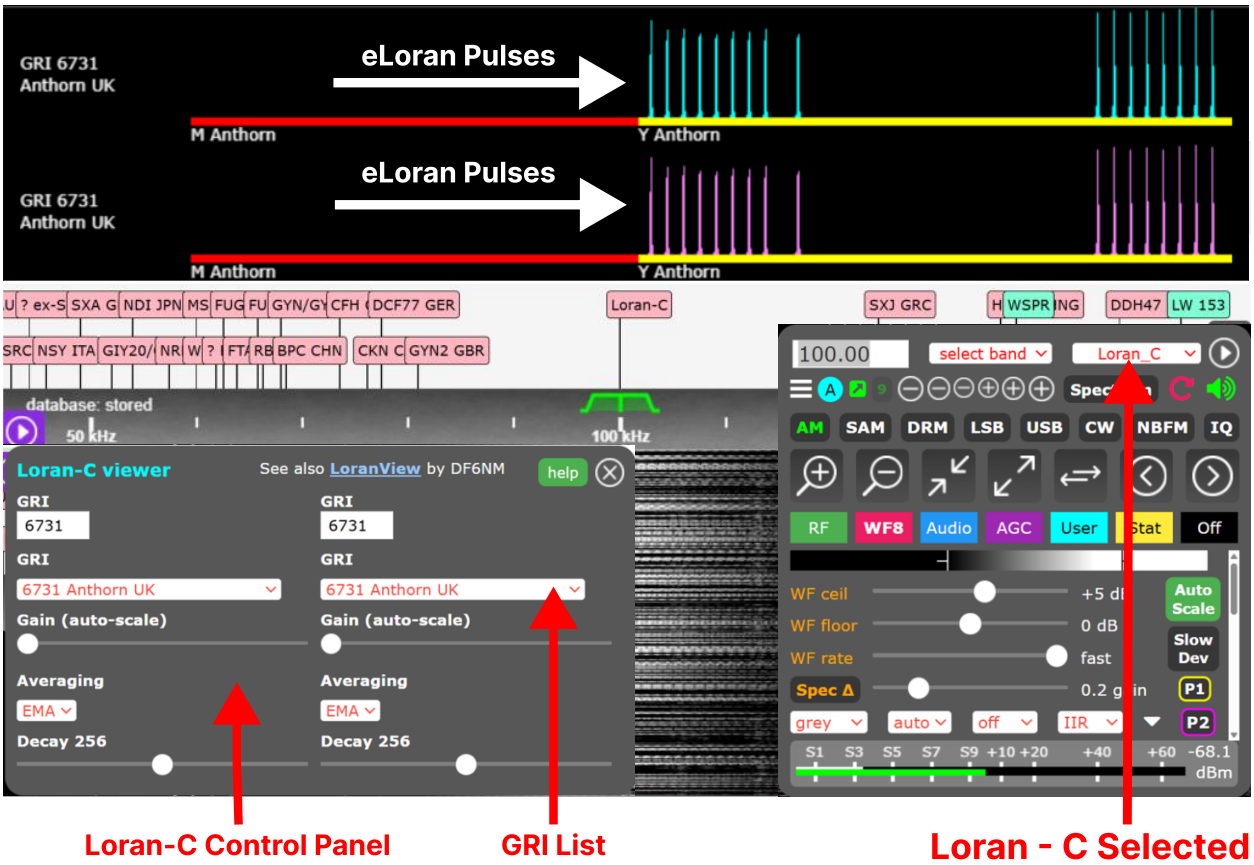
    *Note: the reflection here is based on bit concatenation. We discovered later in the final stages that if the bits are concatenated with the newest to the left, the reflections of the payload and checksum will not be the case.*

The results will now be discussed as case studies: the UK, the Kingdom of Saudi Arabia, the Republic of South Korea, and China.

## 3.1 The United Kingdom – GRI: 6731

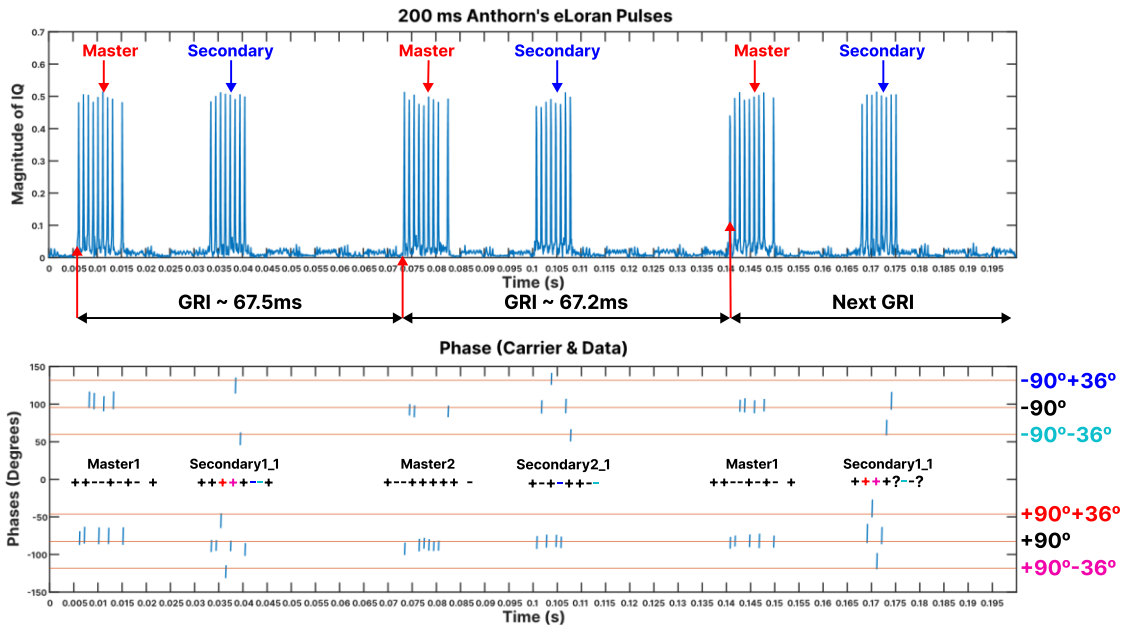
The *eLoran* station in the UK is located at Anthorn, Cumbria. As it would be, there are plenty of Kiwi receivers near Anthorn that can receive the Loran-C signal (they have the Loran-C extension). Below is a figure showing the pulses from Anthorn on a kiwi receiver's webpage. Observations indicate that the pulses originate from the same station, suggesting that both the master and secondary pulses are from a single transmitter.

**Figure 17.** *eLoran* pulses from Anthon on the KiwiSDR webpage




The signal was recorded as the file ***20251014T122009Z_100000_G0GHK_iq.wav*** and samples extracted using the Kiwi framework. We carry out the phase extraction processing, soft to hard bit conversion, and bit concatenation. Below is a figure showing the pulses and phases of the Anthorn signal, indicating master and secondary pulse groups, GRI ~ 67.5 ms, carrier, and data phase modulation. In some cases, we lose the phases either due to the low sampling frequency of the receiver or interference.

**Figure 18.** Pulses and phases of Anthorn's *eLoran* signal.




### 3.1.1 Payload analysis for Anthorn - *20251014T122009Z_100000_G0GHK_iq.wav*

Before the analysis, the following facts are used to validate our interpretations of the payload.

- Hour of the Year: this file was recorded on the 14th of October, 2025, at 5 pm. According to this online calculator, the hour of the year = **6876** hours.
- Year: The recording took place in the year **2025**.
- Leap seconds (TAI-UTC): Loran-C, Long Range Navigation time, is an atomic time scale implemented by the atomic clocks in Loran-C chain transmitter sites. Loran time was zero at 0h 1-Jan-1958, and since it is not perturbed by leap seconds, it is now ahead of UTC by **27** seconds (Leapsecond.com, last accessed February, 2026).
- Station ID: **549** (Time Nut mailing list)
- Station Longitude: **-3.28764** and Latitude: **54.9114** (Open Street Map).

We will now analyse some messages. These messages are taken from crc_eloran.txt. Note that we will only focus on message types 4 (Station ID/Health Message) and 6 (Loran UTC Message) as these are the only messages verifiable with publicly available data. Each payload components are separated and reflected before conversion, based on the message type.

**Message at Bit 0015**; Data; 011010011111010101100101101100111000011101101011010011 00

**0110** **10** **01111101010110010110110011100** **00111011010110** **100110** **0**

**0110** – Flipped to 0110 = Message type 6 – Loran UTC Message.

**10** – Flipped to 01 = Message sub-type 1

**01111101010110010110110011100** – Bits flipped. Master/Secondary Time (unit 10us) = 121019070

**00111011010110** – Bits flipped. Hour of year = 6876

**100110** – Bits Flipped. Year = 25

**0** – Spare.

**Message at Bit 0225**; Data; 011001000100011111010110011100111000000000000011011000000

**0110** **01** **00010001111101011001110011100** **0000000000** **110110000** **00**

**0110** – Flipped to 0110 = Message type 6 – Loran UTC Message.

**01** – Flipped to 10 = Message sub-type 2

**00010001111101011001110011100** – Bits flipped. Master/Secondary Time (unit 10us) = 121221000

**0000000000** – Precise time (unit 10ns) = 0

**110110000** – Bits flipped. Leap seconds (unit 1s) = 27

**00** – Leap seconds change = 0

**Message at Bit 0435**; Data; 011010010010100010001100111100111000011101101011010011 00

**0110** **10** **01001010001000110011110011100** **00111011010110** **100110** **0**

**0110** – Flipped to 0110 = Message type 6 – Loran UTC Message.

**10** – Flipped to 01 = Message sub-type 1

**01001010001000110011110011100** – Bits flipped. Master/Secondary Time (unit 10us) = 121422930

**00111011010110** – Bits flipped. Hour of year = 6876

**100110** – Bits Flipped. Year = 25

**0** – Spare.

**Message at Bit 3375**; Data; 001010100100011111000101000110010001101001010000011111 11

**0010** **1010010001** **111** **10** **001** **01** **00011001000110100101000001111111**

**0010** – Flipped to 0100 = Message type 4 - Eurofix Station ID/Health message

**1010010001** – Bits flipped. Reference Station ID = 549

**111** – Bits flipped. Reference Station Health = Reference Station Not Working

**10** – Bits flipped. System Indicator = *eLoran*

**001** – Bits flipped. Master/Secondary = Yankee Secondary

**01** – Bits flipped. Subtype = GPS Ref Longitude

**00011001000110100101000001111111** – Bits flipped (units 0.0000001 degrees) = 4.2621e+09

**Message at Bit 9682**; Data; 001010100100011111000110100011110111001101011101000001 00

**0010** **1010010001** **111** **10** **001** **10** **10001111011100110101110100000100**

**0010** – Flipped to 0100 = Message type 4 - Eurofix Station ID/Health message

**1010010001** – Bits flipped. Reference Station ID = 549

**111** – Bits flipped. Reference Station Health = Reference Station Not Working

**10** – Bits flipped. System Indicator = *eLoran*

**001** – Bits flipped. Master/Secondary = Yankee Secondary

**10** – Bits flipped. Subtype = GPS Ref Latitude

**10001111011100110101110100000100** – Bits flipped (units 0.0000001 degrees) = 549113585

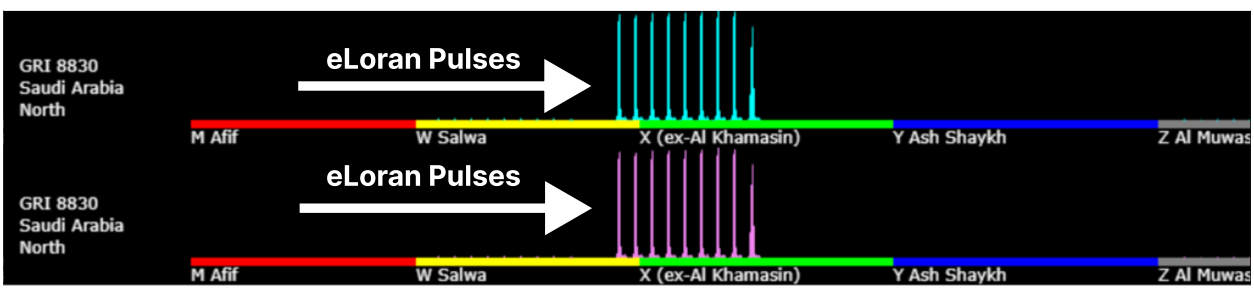
**Message at Bit 15989;** Data; 001010100100011111000101000110010001101001010000011111111
**0010** **1010010001** **111** **10** **001** **01** **0001100100011010010100000111111**
**0010** – Flipped to 0100 = Message type 4 - Eurofix Station ID/Health message
**1010010001** – Bits flipped. Reference Station ID = 549
**111** – Bits flipped. Reference Station Health = Reference Station Not Working
**10** – Bits flipped. System Indicator = *eLoran*
**001** – Bits flipped. Master/Secondary = Yankee Secondary
**01** – Bits flipped. Subtype = GPS Ref Longitude
**0001100100011010010100000111111** – Bits flipped (units 0.0000001 degrees) = 4.2621e+09

From the above brief analysis, we see that the results match the publicly available data. Other message types were present; however, these are differential corrections to GPS, GLONASS, and *eLoran* Phase, Almanac information. To analyse these other types of messages, other systems and references would need to be established. We are quite satisfied with the results above.

### 3.2 The Kingdom of Saudi Arabia – GRI: 8830

The eastern region of Saudi Arabia hosts the *eLoran* station in Salwa with a transmission power of 1000kW. We are lucky to have only one Kiwi receiver in that region, which can receive and process Loran-C signals. A figure showing the pulses from Salwa station as seen on the Kiwi receiver is shown below.

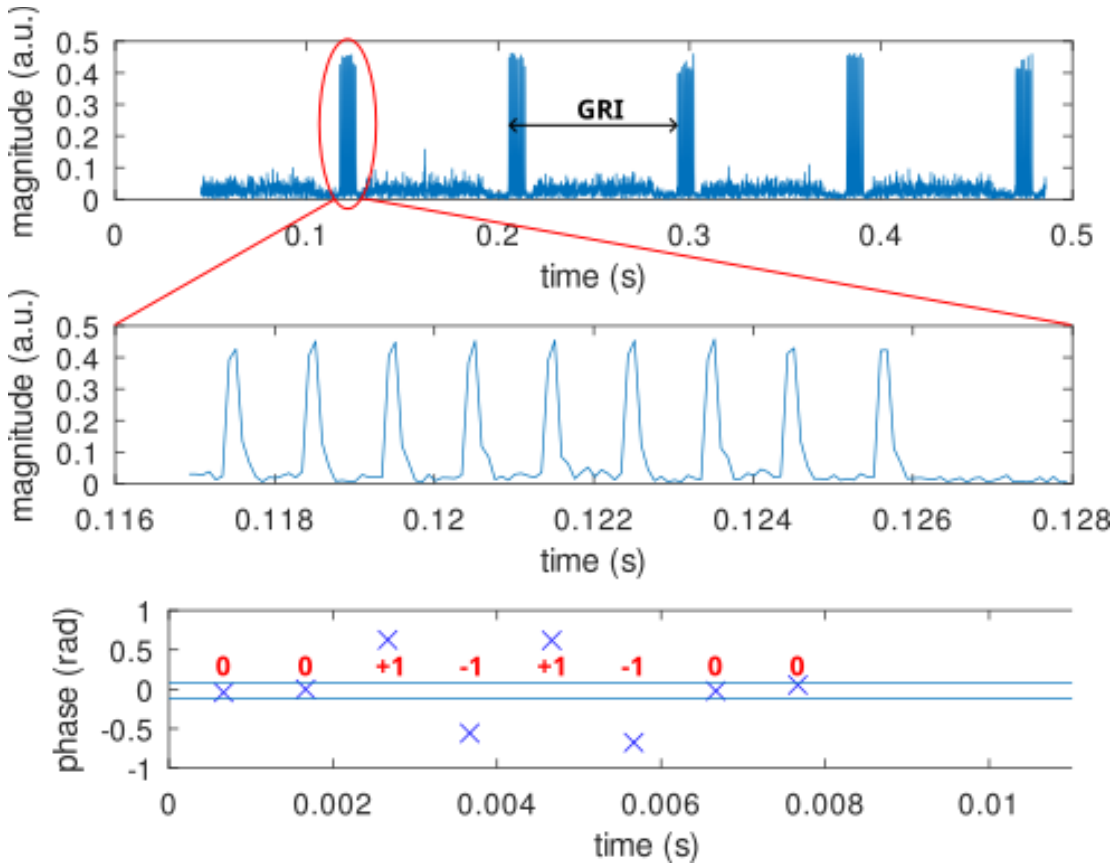
**Figure 19.** Salwa *eLoran* pulses on the KiwiSDR webpage.




As shown above, there are quite a few peculiarities with the Saudi transmissions. First, we see that only one pulse group is broadcast throughout the GRI, which indicates that only one station is operational (Salwa). In addition, instead of 8 pulses (for secondary) or 9 pulses (pulses 8 and 9 spaced by 2 ms for master), we see 9 pulses with an even 1 ms spacing. We had thought that this signal implements the 9th pulse PPM; however, upon investigation by discarding the 9th pulse and applying our already established Eurofix algorithm, lo and behold, we could analyse the payload.

A stream was recorded, named ***20260107T144245Z_100000_DOHA_iq.wav***. It was further processed with the algorithms, phases were extracted, bits were concatenated, the start/stop of messages were found, and corrections were applied. See the figure below for the phase.

**Figure 20.** Phase plot for Salwa, Saudi Arabia




### 3.2.1 Payload analysis for Salwa - *20260107T144245Z_100000_DOHA_iq.wav*

We provide the information below for analysis validation. Message types 4 (Station ID/Health Message) and 6 (Loran UTC message) are selected for analysis.

- Hour of the Year: this file was recorded on the 7th of January, 2026, at 2 pm. According to this online calculator, the hour of the year = **158** hours.
- Year: the recording took place in the year **2026**.
- Leap seconds (TAI-UTC): Loran-C, Long Range Navigation time, is an atomic time scale implemented by the atomic clocks in Loran-C chain transmitter sites. Loran time was zero at 0h 1-Jan-1958, and since it is not perturbed by leap seconds, it is now ahead of UTC by **27** seconds (Leapsecond.com, last accessed February, 2026).
- Station Longitude: **50.5701590** and Latitude: **24.83375**

**Message at Bit 99;** Data; 001000011111000001001001011010110010011000100100011111000
**0010** **0001111100** **000** **10** **010** **01** **01101011001001100010010001111000**
**0010** – Flipped to 0100 = Message type 4 - Eurofix Station ID/Health message
**0001111100** – Bits flipped. Reference Station ID = 248
**000** – Bits flipped. Reference Station Health; UDRE Scale Factor = 1
**10** – Bits flipped. System Indicator = *eLoran*
**010** – Bits flipped. Master/Secondary = Whiskey Secondary
**01** – Bits flipped. Subtype = GPS Ref Longitude
**01101011001001100010010001111000** – Bits flipped (units 0.0000001 degrees) = 505701590

**Message at Bit 939;** Data; 011010000001110101010010111010111100111100100000000101100
**0110** **10** **00000111010101001011101011110** **01111001000000** **010110** **0**
**0110** - Flipped to 0110 = Message type 6 – Loran UTC Message.
**10** - Flipped to 01 = Message sub-type 1
**00000111010101001011101011110** - Bits flipped. Master/Secondary Time (unit 10us) = 257764064
**01111001000000** - Bits flipped. Hour of year = 158
**010110** - Bits Flipped. Year = 26
**0** – Spare

**Message at Bit 2619;** Data; 011001000000001000000110111110111100010111110110110000000
**0110** **01** **00000000100000011011111011110** **0010111110** **110110000** **00**
**0110** – Flipped to 0110 = Message type 6 – Loran UTC Message.
**01** – Flipped to 10 = Message sub-type 2
**00000000100000011011111011110** – Bits flipped. Master/Secondary Time (unit 10us) = 259883264
**0010111110** – Precise time (unit 10ns) = 500
**110110000** – Bits flipped. Leap seconds (unit 1s) = 27
**00** – Leap seconds change = 0

**Message at Bit 17529**; Data; 001000011111000001001010001000111010101010110011011110000
**0010** **0001111100** **000** **10** **010** **10** **00100011101010101011001101110000**
**0010** – Flipped to 0100 = Message type 4 - Eurofix Station ID/Health message
**0001111100** – Bits flipped. Reference Station ID = 248
**000** – Bits flipped. Reference Station Health; UDRE Scale Factor = 1
**10** – Bits flipped. System Indicator = *eLoran*
**010** – Bits flipped. Master/Secondary = Whiskey Secondary
**10** – Bits flipped. Subtype = GPS Ref Latitude
**00100011101010101011001101110000** – Bits flipped (units 0.0000001 degrees) = 248337860

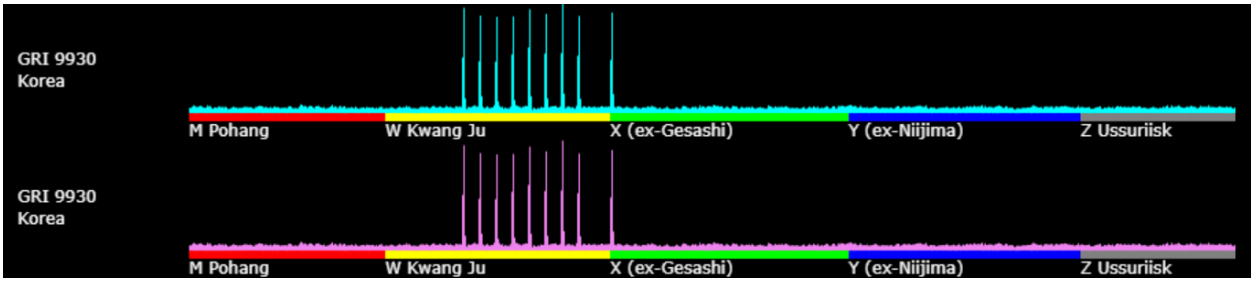
The Saudi interpretations do match with the facts stated above; in addition, 248 as the reference station ID is consistent, hence it can also be taken as the right value.

### 3.3 The Republic of South Korea – GRI: 9930

Since the Koreans are also broadcasting *eLoran*, we also investigated their transmissions to see if they implement Eurofix or 9th pulse modulation, and also if our algorithm is compatible with it. With a Kiwi receiver in Daegu city, we took a record. The signal as seen on the webpage is shown below. What we noticed is a discrepancy between recent publications and what we see on the receiver's webpage. Recent Korean-based publications, as of 2025, highlight that there are three active stations located in Pohang

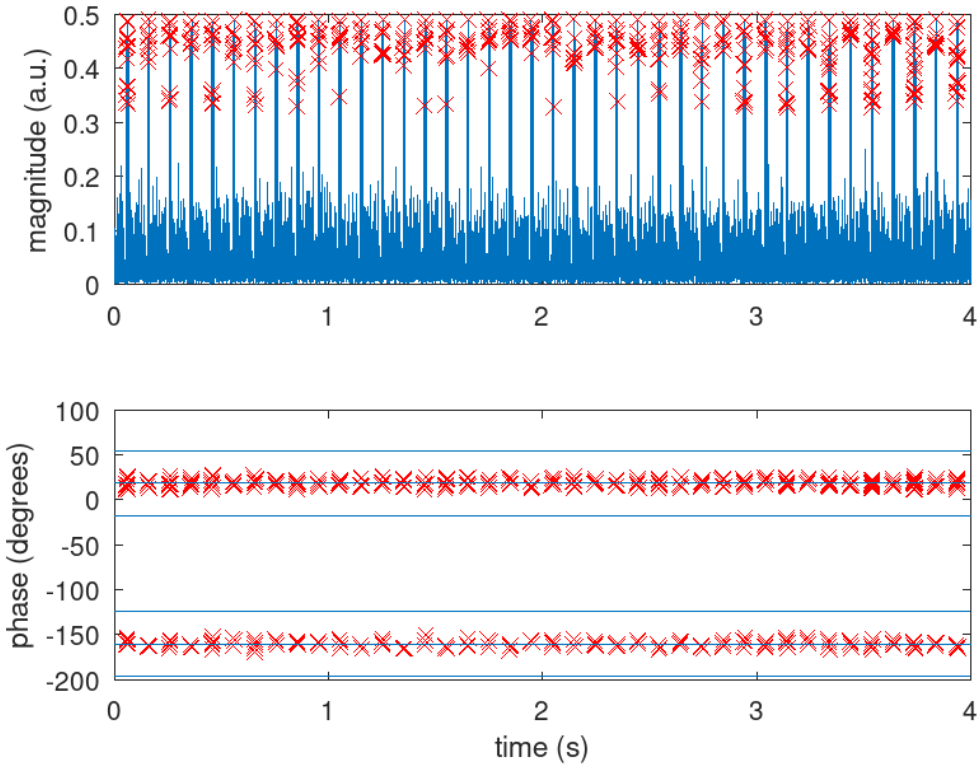
(Master – 9930), Gwangju (Secondary – 9930), and Socheongdo (Secondary – 9930). However, only one pulse group (master pulse group) seems to be received. Frankly, we do not know why.

**Figure 21.** *eLoran* pulses from Korea.




We proceeded and processed the complex data with the algorithm to extract the phases and the bits. The script *process_eloran.m* shows that the phases of the pulses (including 3-8) are only modulated ±90°, forming a "0 0 0 0 0 0" pattern, and there is no ±1 us shift for data modulation. In addition, only master pulse groups were detected in line with the carrier phase codes to form a phase code interval; there were no secondary pulse groups.

**Figure 22.** *eLoran* signal's magnitude and phase for Korea.



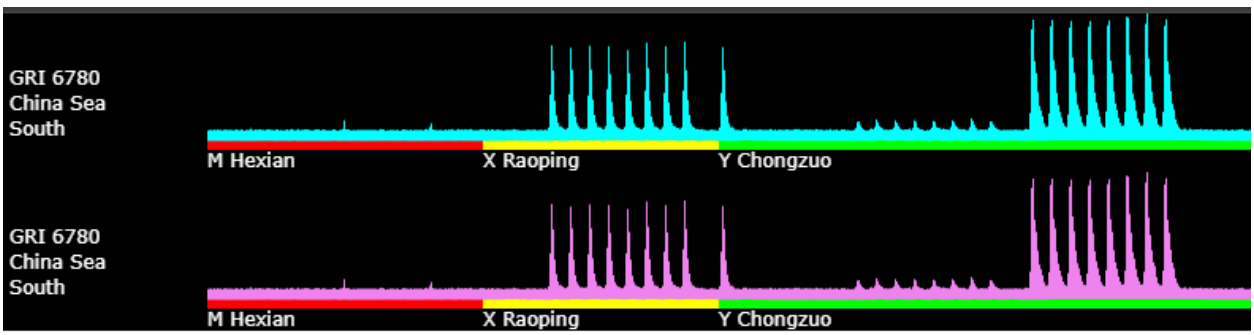

As seen in the ITU-R Recommendation on page 6 (Rec. ITU-R M.589-3, 2001), a blanked pulse group should be considered to have been transmitted for modulation purposes. The document also states that the pattern "0 0 0 0 0 0" should be used to indicate that no data is being transmitted. The case of the Korean

pulses is one of these two, closer to a “No data Transmission pattern” than “Blanking”. We ended work on Korea’s signal after this discovery.

### 3.4 The South Chain – China – GRI: 6780

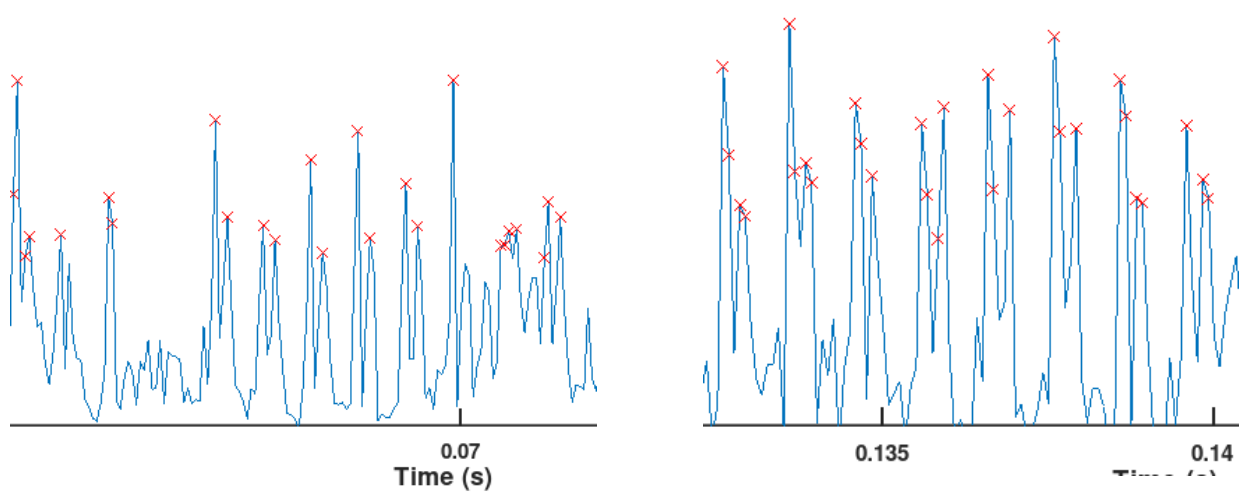
The Chinese do have plenty of Loran chains (6000, 7430, 8390, 6780). We had hoped to examine all chains; however, there are no Kiwi receivers in China. We observed these four chains on Kiwis around China; only one of them, in Hanoi, Vietnam, was able to receive the pulses from the south chain (6780), which we further looked into. The figure below shows the south chain’s pulses as seen on the Hanoi receiver.

**Figure 23.** China’s south chain pulses on KiwiSDR in Vietnam.




From the figure above, we see three stations: the Master and two secondaries; however, the first secondary station’s pulses are so weak, very close to the noise level. This is likely due to low transmitting power from the station or the distance between the station and the Vietnam-based Kiwi receiver. The signal was recorded, and data was extracted using Kiwi’s framework. We plot a portion of the signal’s magnitude and phases as shown below.

**Figure 24.** South chain signal magnitude, phases, and exploded view.



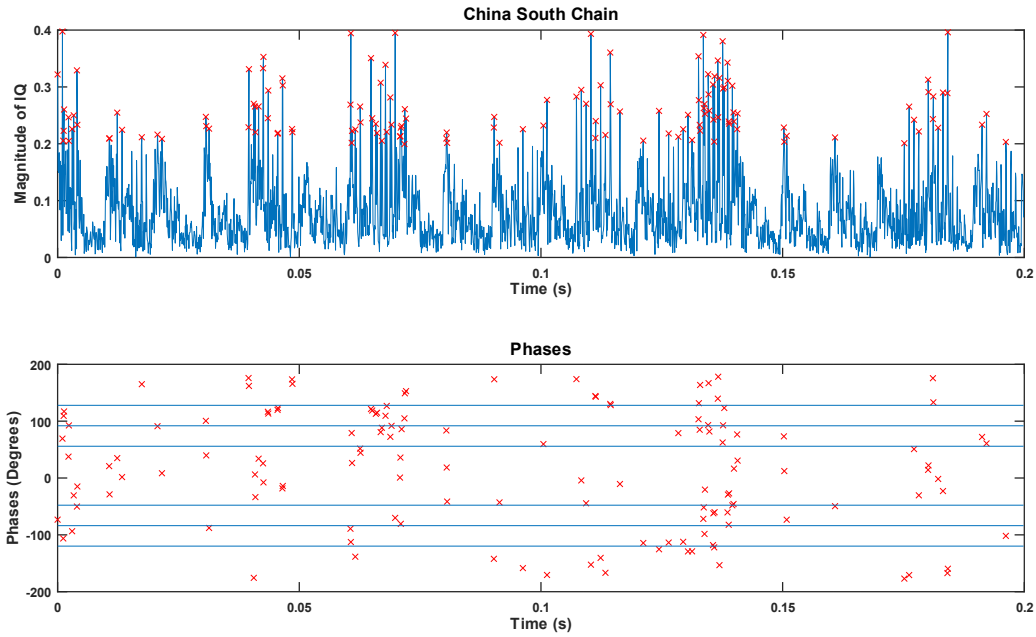
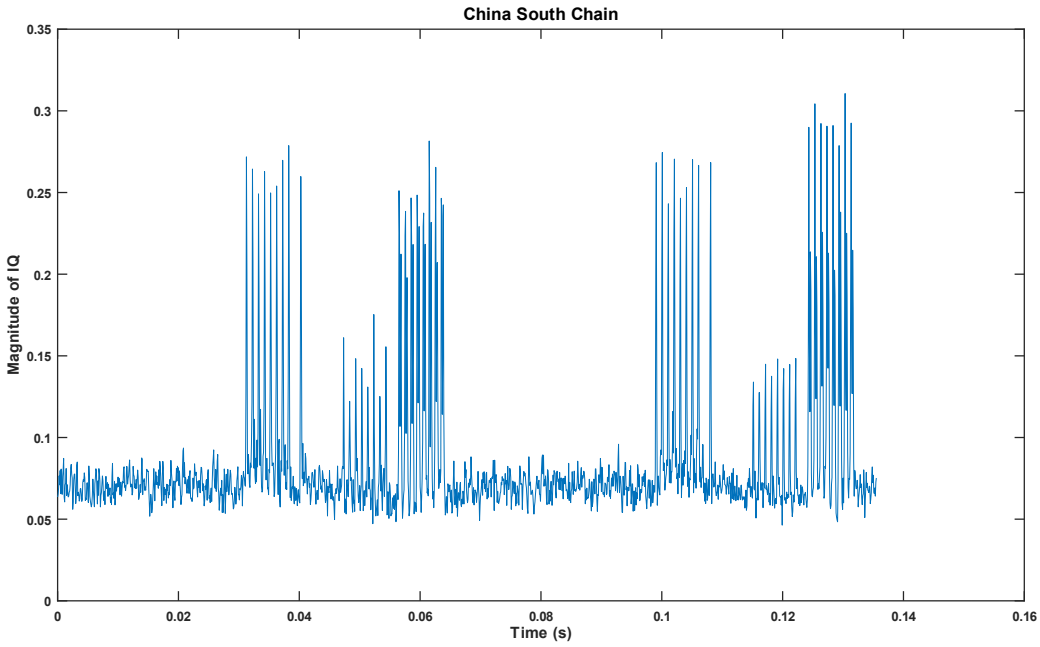
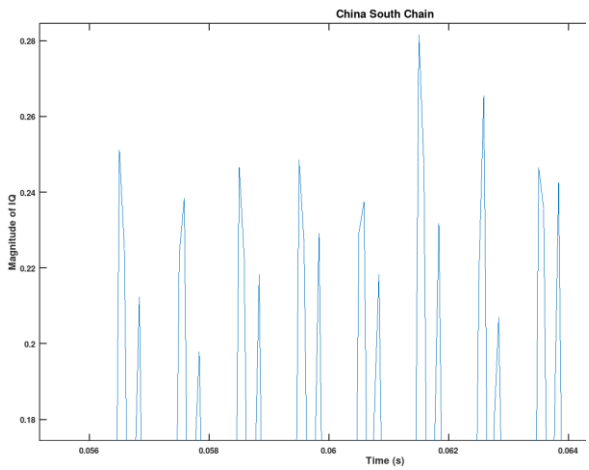
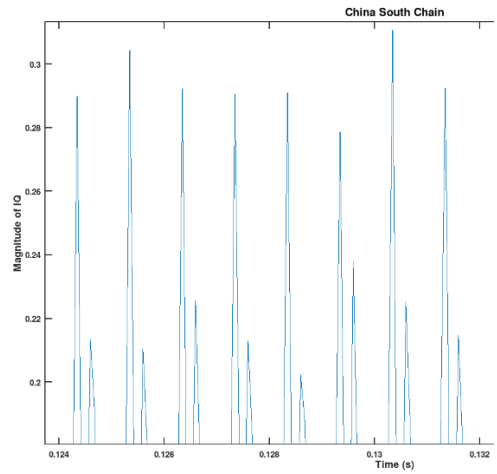



**Figure 25.** Signal's averaging with exploded view






The plot shows exactly what we saw already on the receiver's webpage; in addition, we see some irregularities in the pulses. Two pulse peaks are seen on the last secondary pulse groups; we do not know if this is a receiver-related issue, if it's a transmission issue, or if there is some interference from another Chinese chain. In any case, this was somewhat problematic; it was quite difficult to know the actual pulse and extract the right phase.

We worked on extracting the phases by implementing a combination of different techniques, including a matched filter (pulse group template), pulse position approximation using the strongest pulse, emission delays and GRI, coherent group detection that sums up the energy at the 8 expected pulse times, and checks if it exceeds noise. All these methods proved abortive, and work was put on hold on this chain.

Although not reported here, we might also check the signal from the Russians and the US.

## Chapter 4 – Conclusions

Low-frequency *eLoran*, birthed from Loran-C, is an internationally standardised positioning, navigation, and time service system. It utilises solid-state transmitters, data links for payload, integrity monitoring, differential enhancement, and digital reception. *eLoran* is a pulsed signal characterised by a stable phase and precise prediction of ground wave propagation to achieve high reliability, accuracy, availability, integrity, and continuity performance requirements for

- Aviation non-precision instrument approaches
- Maritime harbour entrance and approach manoeuvres
- Land-mobile vehicle navigation
- Location-based services
- Users of precise time and frequency.

*eLoran* is independent, dissimilar, and complementary to GNSS. It can serve as a backup if satellite services are disrupted/unavailable. It does not share the vulnerability of GNSS to incidental or deliberate jamming, intentional spoofing, radio-frequency interference, or space weather events. Several countries, including France, are actively looking to integrate, collaborate, and form a consortium for *eLoran*. *eLoran* overcomes Loran-C's additional secondary factor by providing differential correction messages. Although its receivers are designed to reject sky-waves as they arrive much later than ground-waves, local consumer devices indoors can interfere with the signal.

KiwiSDR, a widely available and free system, was utilised as an acquisition source for this work. Kiwi provides a framework to extract the complex data from a recording. The data acquired with this radio were suitable to produce the results we had. Despite its resourcefulness, low sampling frequency, weak, noisy signals, and distance from transmitters were limitations experienced during the course of this work.

Despite the inadequacies of the standard, this work has been able to establish soft decoding and deciphering of *eLoran*'s Eurofix modulation. The advanced pulse represented as (-, -1) is +36° ahead, and the delayed pulse represented as (+, +1) is -36° behind, both in reference to the prompt pulses (unmodulated, indicated as 0). The concatenation pattern of bits determines reflection for CRC computation. *Libfec*, an already existing utility, was used to correct message symbols. The least significant symbol of an FEC message is transmitted first; this means that parity symbols are transmitted before the payload.

Towards the end of this work, we discovered that the Saudi Arabians are combining Eurofix (pulses 3-8) and 9th pulse modulation (pulse 9) on a pulse group. This method is feasible as both techniques exist independently and do not interfere with one another. We cannot promise to work on the 9th pulse modulation due to the low sample rate of the Kiwi receivers; however, be on the lookout for the GitHub repository.

We were unable to ascertain something I would call the mystery around the Korean transmissions; Kiwi receivers have worked so well (when close enough to *eLoran* stations) that we can rely on them to examine and record. Although Eurofix Architect, Gerald Offerman, did mention that the Koreans do implement Eurofix, this negates what we found. We suspect that data is unmodulated onto the master signals, just like in Anthorn; if this is a standard, we do not know. Another point could be around the Kiwi receivers because they are tuned for high-frequency signal reception and not for a 100kHz *eLoran* signal.

Due to the absence of Kiwi receivers in China, this work could not investigate the Chinese *eLoran* system. Based on publications, the Chinese are very active with *eLoran*; they have incorporated *eLoran* and fibre optics to provide reliable timing information.

More work around the acquisition of *eLoran* signals, robust signal processing techniques, and 9th pulse modulation could be looked into.

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*SAE INTERNATIONAL SAE9990™/2 Page 2 of 1 3 GNSS Global Navigation Satellite System GPS Global Positioning System GRI Group Repetition Interval LAT Latitude LF Low Frequency LON Longitude LORAN Long-Range Navigation PCI Phase-Code Interval PNT Positioning, Navigation, and Timing RS Reed-Solomon TDMA Time Division Multiple Access*. (n.d.).

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**Appendix**

**Pattern/Data Translation (ITU, 2001).**

## Pattern/data translation

| Decimal | Hexa-decimal | Pattern | Decimal | Hexa-decimal | Pattern | Decimal | Hexa-decimal | Pattern |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | --00++ | 43 | 2B | 00-+-+ | 86 | 56 | ++-00- |
| 1 | 1 | --0+0+ | 44 | 2C | 00-++- | 87 | 57 | ++0--0 |
| 2 | 2 | --0++0 | 45 | 2D | 00+--+ | 88 | 58 | ++0-0- |
| 3 | 3 | --+00+ | 46 | 2E | 00+-+- | 89 | 59 | ++00-- |
| 4 | 4 | --+0+0 | 47 | 2F | 00++-- | 90 | 5A | -0000+ |
| 5 | 5 | --++00 | 48 | 30 | 0+--0+ | 91 | 5B | -000+0 |
| 6 | 6 | -0-0++ | 49 | 31 | 0+--+0 | 92 | 5C | -00+00 |
| 7 | 7 | -0-+0+ | 50 | 32 | 0+-0-+ | 93 | 5D | -0+000 |
| 8 | 8 | -0-++0 | 51 | 33 | 0+-0+- | 94 | 5E | -+0000 |
| 9 | 9 | -00-++ | 52 | 34 | 0+-+-0 | 95 | 5F | 0-000+ |
| 10 | A | -00+-+ | 53 | 35 | 0+-+0- | 96 | 60 | 0-00+0 |
| 11 | B | -00++- | 54 | 36 | 0+0--+ | 97 | 61 | 0-0+00 |
| 12 | C | -0+-0+ | 55 | 37 | 0+0-+- | 98 | 62 | 0-+000 |
| 13 | D | -0+-+0 | 56 | 38 | 0+0+-- | 99 | 63 | 00-00+ |
| 14 | E | -0+0-+ | 57 | 39 | 0++--0 | 100 | 64 | 00-0+0 |
| 15 | F | -0+0+- | 58 | 3A | 0++-0- | 101 | 65 | 00-+00 |
| 16 | 10 | -0++-0 | 59 | 3B | 0++0-- | 102 | 66 | 000-0+ |
| 17 | 11 | -0++0- | 60 | 3C | +--00+ | 103 | 67 | 000-+0 |
| 18 | 12 | -+-00+ | 61 | 3D | +--0+0 | 104 | 68 | 0000-+ |
| 19 | 13 | -+-0+0 | 62 | 3E | +--+00 | 105 | 69 | 0000+- |
| 20 | 14 | -+-+00 | 63 | 3F | +-0-0+ | 106 | 6A | 000+-0 |
| 21 | 15 | -+0-0+ | 64 | 40 | +-0-+0 | 107 | 6B | 000+0- |
| 22 | 16 | -+0-+0 | 65 | 41 | +-00-+ | 108 | 6C | 00+-00 |
| 23 | 17 | -+00-+ | 66 | 42 | +-00+- | 109 | 6D | 00+0-0 |
| 24 | 18 | -+00+- | 67 | 43 | +-0+-0 | 110 | 6E | 00+00- |
| 25 | 19 | -+0+-0 | 68 | 44 | +-0+0- | 111 | 6F | 0+-000 |
| 26 | 1A | -+0+0- | 69 | 45 | +-+-00 | 112 | 70 | 0+0-00 |
| 27 | 1B | -++-00 | 70 | 46 | +-+0-0 | 113 | 71 | 0+00-0 |
| 28 | 1C | -++0-0 | 71 | 47 | +-+00- | 114 | 72 | 0+000- |
| 29 | 1D | -++00- | 72 | 48 | +0--0+ | 115 | 73 | +-0000 |
| 30 | 1E | 0--0++ | 73 | 49 | +0--+0 | 116 | 74 | +0-000 |
| 31 | 1F | 0--+0+ | 74 | 4A | +0-0-+ | 117 | 75 | +00-00 |
| 32 | 20 | 0--++0 | 75 | 4B | +0-0+- | 118 | 76 | +000-0 |
| 33 | 21 | 0-0-++ | 76 | 4C | +0-+-0 | 119 | 77 | +-+-+- |
| 34 | 22 | 0-0+-+ | 77 | 4D | +0-+0- | 120 | 78 | -+-+-+ |
| 35 | 23 | 0-0++- | 78 | 4E | +00--+ | 121 | 79 | +-+--+ |
| 36 | 24 | 0-+-0+ | 79 | 4F | +00-+- | 122 | 7A | -+-++- |
| 37 | 25 | 0-+-+0 | 80 | 50 | +00+-- | 123 | 7B | +--+-+ |
| 38 | 26 | 0-+0-+ | 81 | 51 | +0+--0 | 124 | 7C | -++-+- |
| 39 | 27 | 0-+0+- | 82 | 52 | +0+-0- | 125 | 7D | +--++- |
| 40 | 28 | 0-++-0 | 83 | 53 | +0+0-- | 126 | 7E | -++--+ |
| 41 | 29 | 0-++0- | 84 | 54 | ++--00 | 127 | 7F | +0000- |
| 42 | 2A | 00--++ | 85 | 55 | ++-0-0 | | | |

**General Eurofix Message Format (Offermans, 2014).**

| | |
|---|---|
| Application Data | 70 bits |
| Reed-Solomon Forward Error Correction | 140 bits |
| Total | 210 bits |

**Application Data General Bit Allocation (Offermans, 2014).**

| | |
|---|---|
| Message Type | 4 bits |
| Message Content | 52 bits |
| Cyclic Redundancy Check | 14 bits |
| Total | 70 bits |

**Eurofix Message Description and Status (Offermans, 2014).**

Fixed - Standardised by ITU, or implemented in operational installations.

Tentative - Implemented and operational in *eLoran* installations, not standardised.

Obsolete - Implemented for trial, not operational anymore.

Reserved - Free for future use.

| Message Type | Bits | Description | Status |
|---|---|---|---|
| 1 | 0001 | Eurofix DGPS correction (single sat) | Fixed |
| 2 | 0010 | Eurofix DGlonass correction (single sat) | Fixed |
| 3 | 0011 | Reserved | |
| 4 | 0100 | Eurofix Station ID/Health message | Fixed |
| 5 | 0101 | Short Message Service (SMS) | Fixed |
| 6 | 0110 | Loran UTC Message | Fixed |
| 7 | 0111 | Reserved | |
| 8 | 1000 | Reserved | |
| 9 | 1001 | Reserved | |
| 10 | 1010 | Differential eLoran Phase Corrections | Fixed |
| 11 | 1011 | Differential eLoran Almanac data | Tentative |
| 12 | 1100 | Third Party Data Client | Fixed |
| 13 | 1101 | Reserved | |
| 14 | 1110 | Reserved | |
| 15 | 1111 | Reserved | |
| 16 | 0000 | Reserved | |

**9th Pulse Modulation; Symbol delays from Zero-symbol Offset (SAE 9990-2).**

The 5 MHz clock rate reflected below need not be viewed as a limiting implementation technique.

| i = [0, 7] | | i = [8, 15] | | i = [16, 23] | | i = [24, 31] | |
|---|---|---|---|---|---|---|---|
| **0** | 0.0 | **8** | 50.6 | **16** | 101.2 | **24** | 151.8 |
| **1** | 1.2 | **9** | 51.8 | **17** | 102.6 | **25** | 153.2 |
| **2** | 2.6 | **10** | 53.2 | **18** | 103.8 | **26** | 154.4 |
| **3** | 3.8 | **11** | 54.4 | **19** | 105.0 | **27** | 155.6 |
| **4** | 5.0 | **12** | 55.6 | **20** | 106.2 | **28** | 156.8 |
| **5** | 6.2 | **13** | 56.8 | **21** | 107.6 | **29** | 158.2 |
| **6** | 7.6 | **14** | 58.2 | **22** | 108.8 | **30** | 159.4 |
| **7** | 8.8 | **15** | 59.4 | **23** | 110.0 | **31** | 160.6 |

**9th Pulse Modulation; Message Components (SAE 9990-2).**

| Section | Type | Payload | Parity |
|---|---|---|---|
| **Length (bits)** | 4 | 41 | 75 |
| **Bit Assignment** | 0…3 | 4…44 | 45…119 |

**9th Pulse Modulation; Message Types (SAE 9990-2).**

| Number | Type code | Description |
|---|---|---|
| 0 | 0000 | Reference Site Phase Correction |
| 1 | 0001 | Almanac (further divided into sub-messages) |
| 2 | 0010 | Message for government use-only |
| 3 | 0011 | Message for government use only |
| 4 - 14 | 0100 thru 1110 | Undefined |
| 15 | 1111 | Station Identification and Time-of-Day |